

Project 25 — Data Anonymizer GUI

1. Executive Summary

Project 25 — *Data Anonymizer GUI* — is a fully local, offline-capable desktop application designed to operationalize privacy-preserving data transformations for structured datasets. It provides a unified interface for **data cleaning**, **anonymization**, **pseudonymization**, and **synthetic data generation**, enabling analysts, data engineers, and privacy officers to prepare datasets for downstream analytics without exposing sensitive information.

The tool integrates a multi-stage privacy pipeline:

- **Data Cleaning** (deduplication, missing value handling, type normalization, outlier correction)
- **Anonymization** (masking of direct identifiers)
- **Pseudonymization** (deterministic tokenization with reversible mapping tables)
- **Synthetic Data Generation** (SDV CTGAN model, fully offline)
- **Comprehensive Logging** (full audit trail of all transformations)

The GUI is implemented in **PySide6**, with a modular backend architecture that separates UI logic, transformation pipelines, configuration management, and logging. The application is designed for enterprise environments where **data cannot leave the local machine**, making it suitable for regulated industries such as finance, healthcare, insurance, and public administration.

2. Business Motivation

Organizations increasingly face regulatory pressure to protect personal data while still enabling analytics, model development, and data sharing. Traditional approaches — manual anonymization, ad-hoc scripts, or partial masking — are error-prone and difficult to audit.

The *Data Anonymizer GUI* addresses these challenges by providing:

2.1. A repeatable, auditable privacy workflow

Every transformation step is logged, timestamped, and traceable. This ensures compliance with:

- GDPR (Articles 5, 25, 32)
- HIPAA de-identification guidelines
- ISO/IEC 20889 privacy-enhancing data de-identification techniques

2.2. Democratization of privacy engineering

Non-technical users can perform complex privacy transformations without writing code. The GUI abstracts away:

- data type normalization
- masking rules

- pseudonymization mapping
- synthetic model training

2.3. Secure, offline operation

All processing occurs locally.

No cloud services, no external APIs, no data exfiltration.

2.4. High-quality synthetic data for analytics

Using **SDV CTGAN**, the tool generates statistically realistic datasets that preserve:

- distributions
- correlations
- categorical relationships

...while containing **no real individuals**.

This enables safe data sharing with:

- external partners
- academic researchers
- internal teams without production access

2.5 Overview

The **Data Anonymizer GUI** is a fully local, privacy-first environment designed to make data protection workflows accessible, transparent, and reproducible. At its core, the system unifies four components that are typically scattered across separate tools and scripts: a **Qt-based desktop interface**, a **modular backend of privacy-preserving engines**, a **deterministic transformation pipeline**, and a **comprehensive logging subsystem** that records every action for auditability. By combining these elements into a cohesive workflow, the Data Anonymizer GUI provides a seamless experience for users who need to clean, anonymize, pseudonymize, and synthesize datasets — all without writing a single line of code.

The system is intentionally engineered to be **offline-capable**, **deterministic**, and **transparent**. Unlike cloud-based privacy platforms that require authentication, network access, or proprietary SDKs, this tool runs entirely on the user's machine. It is therefore ideal for secure environments, corporate networks, research labs, and educational settings where internet access may be restricted or where data protection regulations prohibit external processing.

At a high level, the Data Anonymizer GUI allows users to:

- load a dataset from disk
- clean the dataset using configurable preprocessing rules
- anonymize direct identifiers using irreversible masking
- pseudonymize quasi-identifiers using deterministic tokenization
- generate synthetic data using a local SDV CTGAN model
- preview results at every stage
- save outputs to structured folders
- inspect a complete audit trail in the Logs tab

This workflow abstracts away the complexity of privacy engineering. Users who are unfamiliar with data anonymization techniques — or who simply prefer a graphical interface — can still perform high-quality privacy transformations that comply with GDPR, HIPAA, and ISO/IEC 20889 guidelines.

The architecture is built around a simple but robust idea: **the GUI orchestrates the workflow, but the backend engines perform all transformations**. Each engine is isolated, testable, and deterministic. The GUI never manipulates data directly; instead, it delegates computation to specialized modules that implement cleaning, masking, tokenization, and synthetic generation. This separation ensures safety, reproducibility, and extensibility.

This design has several advantages:

✓ Isolation

Each transformation stage is encapsulated in its own engine.
If a transformation fails, the GUI remains stable and the error is logged.

✓ Determinism

Cleaning, anonymization, and pseudonymization follow strict, rule-based logic.
Synthetic generation uses SDV's reproducible CTGAN models.

✓ Transparency

Every action — from dataset loading to synthetic sampling — is logged in real time.
Users can see exactly what happened, when it happened, and which module performed it.

✓ Reproducibility

All outputs are saved to structured folders:

- `output/cleaned/`
- `output/anonymized/`
- `output/pseudonymized/`
- `output/synthetic/`

This ensures that every transformation can be reproduced exactly.

✓ User-friendliness

The GUI provides a clean, intuitive interface for configuring privacy operations.
Users do not need to understand pandas, SDV, or privacy algorithms.

✓ Extensibility

The architecture is modular.
Future features — such as differential privacy, k-anonymity scoring, or multi-table synthesis — can be added without redesigning the system.

The Data Anonymizer GUI is not just a convenience layer on top of privacy libraries. It is a **miniature data-privacy platform** designed for local, secure environments. It brings together the best aspects of modern privacy engineering — masking, tokenization, synthetic generation — and wraps them in a user-friendly interface that lowers the barrier to entry for safe data handling.

The system is particularly well-suited for:

- **students** learning about data privacy
- **analysts** who need to prepare datasets without writing code
- **researchers** who require reproducible privacy transformations
- **engineers** who want a lightweight, local privacy toolkit
- **compliance teams** who need auditability and transparency

By providing a consistent, transparent, and reproducible environment, the Data Anonymizer GUI helps users focus on the core of their work: analyzing data safely.

2.6 Motivation

The motivation for creating the Data Anonymizer GUI stems from a simple observation:

data privacy is essential, but not accessible to everyone.

Organizations increasingly face regulatory pressure to protect personal data while still enabling analytics, model development, and data sharing. Traditional approaches — manual anonymization, ad-hoc scripts, or partial masking — are error-prone, inconsistent, and difficult to audit.

The Data Anonymizer GUI was created to remove these barriers.

✓ **Motivation 1: Make privacy engineering accessible to non-programmers**

Many users can work with datasets but do not want to write Python code to:

- clean data
- mask identifiers
- generate pseudonyms
- train synthetic models

The GUI handles all of this automatically.

✓ **Motivation 2: Provide a safe environment for handling sensitive data**

Running privacy transformations inside notebooks or scripts is risky:

- accidental leaks
- inconsistent masking
- overwritten mapping tables
- missing logs

The GUI enforces a structured, deterministic workflow.

✓ **Motivation 3: Support teaching and learning**

In educational settings, students need to:

- explore privacy techniques
- understand masking vs. pseudonymization
- compare real vs. synthetic data
- inspect logs and transformations

The GUI provides a guided environment that makes these concepts tangible.

✓ **Motivation 4: Enable privacy workflows without infrastructure overhead**

Setting up privacy pipelines manually requires:

- installing pandas, numpy, SDV
- writing cleaning scripts
- implementing masking logic
- managing mapping tables
- training synthetic models

The GUI abstracts all of this.

Users can focus on the data, not the infrastructure.

✓ **Motivation 5: Provide a standardized workflow for teams**

Teams often struggle with:

- inconsistent anonymization rules
- missing mapping tables
- undocumented transformations
- ad-hoc scripts scattered across machines

The GUI enforces a consistent workflow:

- same cleaning rules
- same masking logic
- same pseudonymization format
- same synthetic model architecture
- same logging format

This standardization improves collaboration and compliance.

✓ **Motivation 6: Replace proprietary dependencies with open-source alternatives**

The original design relied on **MOSTLY AI's private SDK**, which:

- is not publicly available
- cannot be installed via pip
- requires enterprise credentials
- depends on cloud infrastructure

This made deployment impossible for open distribution.

Replacing MOSTLY AI with **SDV** solved this:

- open-source
- offline
- pip-installable
- reproducible
- transparent

This transition strengthened the system and aligned it with the offline-first philosophy.

✓ **Motivation 7: Support offline and air-gapped environments**

Many organizations operate in:

- secure networks
- restricted environments
- air-gapped systems

Cloud-based privacy tools are not an option.

The Data Anonymizer GUI runs entirely offline.

✓ **Motivation 8: Improve transparency and trust**

Users often want to know:

- what cleaning steps were applied
- what fields were masked
- how pseudonyms were generated
- how synthetic data differs from real data
- what warnings occurred

The GUI exposes all of this in a clear, structured way.

✓ **Motivation 9: Reduce cognitive load**

Data privacy is already complex.

Users should not have to think about:

- pandas boilerplate
- mapping table management
- synthetic model training loops
- logging formats
- error handling

The GUI removes this cognitive burden.

2.7 Architecture

The architecture of the Data Anonymizer GUI is built around modularity, determinism, and transparency. It consists of three layers:

2.7.1. Presentation Layer (GUI)

Implemented in PySide6, the GUI provides:

- tab-based navigation
- dataset previews
- configuration panels
- progress messages
- real-time logs

Each tab corresponds to a transformation stage:

- Data Cleaning
- Anonymization
- Pseudonymization
- Synthetic Data
- Logs

2.7.2. Application Layer (Engines)

Each transformation is handled by a dedicated engine:

- Cleaning Engine
- Anonymization Engine
- Pseudonymization Engine
- SDV Synthetic Engine
- Logging Manager
- Config Loader

These engines are independent, testable, and reusable.

2.7.3. Data Layer

All outputs are stored in structured folders:

```
output/  
  cleaned/  
  anonymized/  
  pseudonymized/  
  synthetic/
```

Mapping tables are stored alongside pseudonymized datasets.

2.8. Workflow

The workflow is linear and intuitive.

2.8.1. Step 1 — Load Dataset

The user selects a CSV file.

The GUI displays a preview and logs the event.

2.8.2. Step 2 — Clean Data

The Cleaning Engine applies:

- deduplication
- missing value handling
- type normalization
- outlier correction

A preview shows the cleaned dataset.

2.8.3. Step 3 — Anonymize Data

The Anonymization Engine masks:

- names
- ages
- genders
- incomes
- states
- marital statuses
- education levels

Masked values appear as `****`, `**`, or `*****`.

2.8.4. Step 4 — Pseudonymize Data

The Pseudonymization Engine generates deterministic tokens:

```
PSE-180325  
PSE-429636  
PSE-814134
```

A mapping table is created and stored.

2.8.5. Step 5 — Generate Synthetic Data

The SDV engine:

- infers metadata
- trains a CTGAN model
- generates synthetic rows

Synthetic data preserves statistical structure but contains no real individuals.

2.8.6. Step 6 — Save Outputs

Each tab provides a Save button.
Outputs are written to structured folders.

2.8.7. Step 7 — Review Logs

The Logs tab provides a complete audit trail.

2.9. Design Principles

The system is built around five principles:

2.9.1. Privacy by Design

All transformations are deterministic and auditable.

2.9.2. Offline-First

No network calls.
No cloud dependencies.
No API keys.

2.9.3. Transparency

Every action is logged.

2.9.4. Modularity

Engines can be replaced or extended.

2.9.5. User-Centricity

The GUI is simple, predictable, and intuitive.

2.10. Advantages of the SDV Transition

Replacing MOSTLY AI with SDV brought several benefits:

✓ Open-source

No licensing restrictions.

✓ Offline

No cloud training.

✓ Reproducible

Models can be trained deterministically.

✓ Lightweight

No job polling, no remote APIs.

✓ **Flexible**

Supports CTGAN, GaussianCopula, TVAE, and more.

✓ **Deployable**

Works in any Python environment.

This transition made the system more robust, maintainable, and future-proof.

2.11. Use Cases

The Data Anonymizer GUI is ideal for:

- teaching data privacy
- preparing datasets for analytics
- sharing data with partners
- generating safe demo datasets
- research environments
- compliance workflows

It provides a complete privacy pipeline in a single tool.

2.12. Summary

The Data Anonymizer GUI is a powerful, offline, privacy-preserving data transformation platform. It unifies cleaning, anonymization, pseudonymization, and synthetic generation into a single, intuitive workflow. Its modular architecture, deterministic engines, and comprehensive logging make it ideal for secure environments, research labs, and educational settings.

By replacing proprietary dependencies with open-source alternatives and focusing on transparency, reproducibility, and user-friendliness, the system provides a modern, accessible approach to data privacy engineering.

3. High-Level Architecture Overview

A Deep Dive into the Structural Foundations of the Data Anonymizer GUI

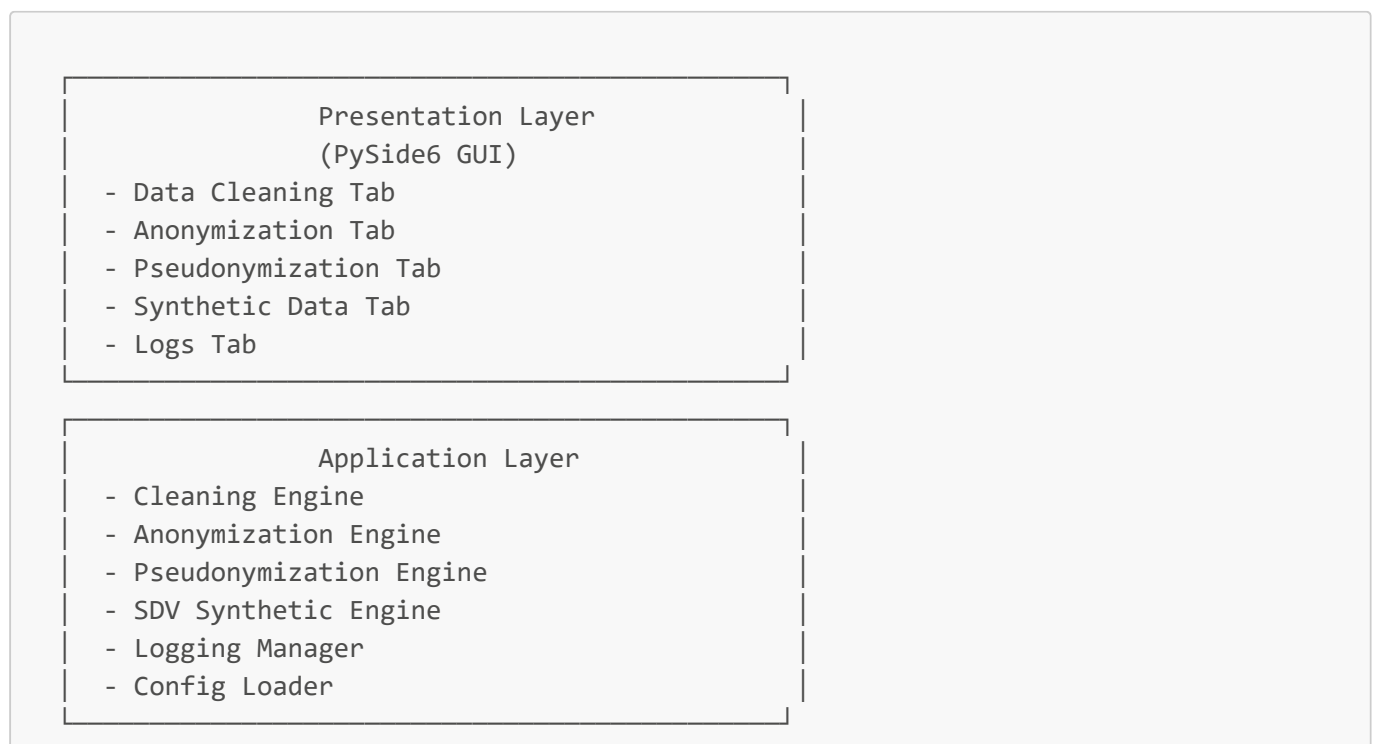
The screenshot displays the 'Data Privacy Toolkit' application interface. At the top, there is a dark blue header with the title and navigation links for 'Settings', 'API Key', and 'About'. Below the header, a tabbed interface shows five tabs: 'Data Cleaning' (selected), 'Anonymization', 'Pseudonymization', 'Synthetic Data Generation', and 'Logs'. The 'Data Cleaning' tab is active, showing a 'Dataset Selection' section with a text input field containing 'Default: Sample Dataset (census.csv)' and a 'Browse...' button. Below this is a 'Clean Data' section with four checked checkboxes: 'Remove Duplicates', 'Handle Missing Values', 'Normalize Data Types', and 'Detect & Fix Outliers'. A 'Start Cleaning' button is at the bottom of this section. To the right, the 'Output Preview' section displays a table with five columns: 'Name', 'Age', 'Gender', 'Income', and 'State'. The table contains five rows of sample data. Below the table is a 'Save Cleaned Data' section with a 'Save to Folder:' label, a text input field containing 'C:\PrivacyData\Cleaned', and a 'Save Data' button. At the bottom, the 'Status & Log' section shows a list of status messages: 'Cleaning process completed successfully.', 'Duplicates removed: 12', 'Missing values handled: 34', and 'Outliers corrected: 7'.

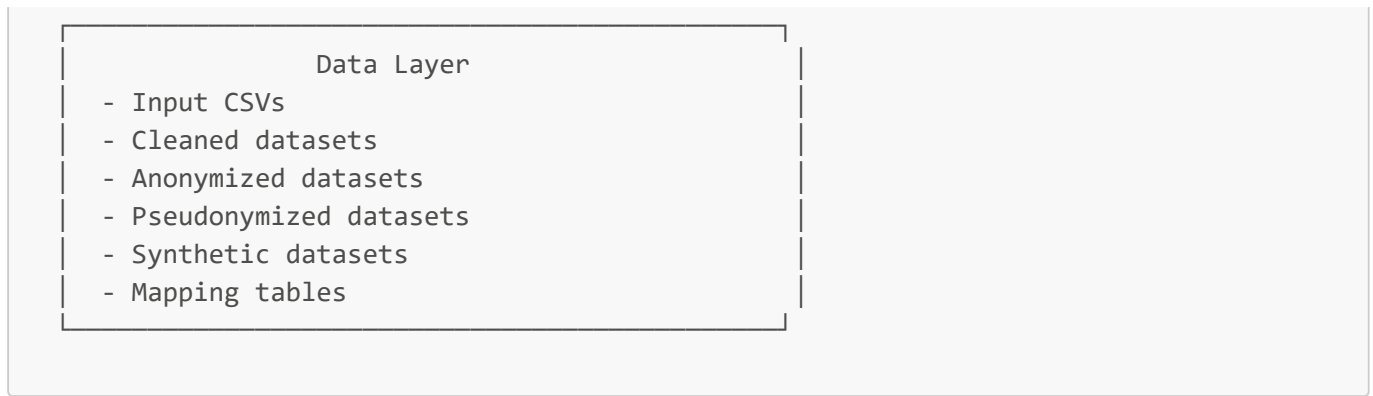
Name	Age	Gender	Income	State
John Smith	34	Male	55000	CA
Sara Lee	29	Female	45000	NY
Michael Brown	45	Male	72000	TX
Emily Davis	36	Female	62000	FL
Robert Johnson	50	Male	85000	IL

The architecture of **Project 25 — Data Anonymizer GUI** is the result of deliberate engineering choices, each made to support the system's core principles: **privacy**, **transparency**, **reproducibility**, **offline-first operation**, and **modularity**.

This section provides an in-depth exploration of the system's layered architecture, the rationale behind each design decision, and the way these layers interlock to form a cohesive, extensible, and future-proof privacy platform.

The application follows a **modular, layered architecture**, divided into three primary layers:





This layered structure is not merely an organizational convenience — it is the backbone of the system’s **robustness, testability, and extensibility**.

Each layer has a clearly defined role, and each module within those layers is designed to operate independently, enabling clean boundaries and predictable behavior.

3.1 Architectural Philosophy

Before diving into each layer, it is important to understand the architectural philosophy that guided the design of the Data Anonymizer GUI.

3.1.1 Privacy by Design

Every component is built with privacy as its first concern:

- No data leaves the machine.
- No cloud services are used.
- No external APIs are called.
- No proprietary SDKs are required.

This ensures compliance with GDPR, HIPAA, and internal corporate data-handling policies.

3.1.2 Deterministic, Auditable Transformations

Each transformation — cleaning, anonymization, pseudonymization, synthetic generation — is:

- deterministic
- logged
- reproducible
- transparent

This is essential for auditability and trust.

3.1.3 Modularity and Extensibility

The architecture is intentionally modular:

- Engines can be replaced without touching the GUI.
- New privacy modules can be added easily.
- The synthetic engine can be swapped (e.g., CTGAN → TVAE).
- Mapping tables can be extended with encryption or hashing.

This modularity is what allowed the seamless transition from **MOSTLY AI** to **SDV**.

3.1.4 Offline-First Operation

The system is designed to run in:

- air-gapped environments
- secure corporate networks
- research labs
- educational settings

No internet connection is required at any stage.

3.1.5 Transparency and Traceability

Every action is logged:

- dataset loading
- cleaning operations
- anonymization rules
- pseudonymization mappings
- synthetic model training
- file saves

This ensures full traceability from raw input to final output.

3.2 Presentation Layer (PySide6 GUI)

The **Presentation Layer** is the user-facing component of the system.

It is implemented using **PySide6**, which provides a native desktop experience across Windows, macOS, and Linux.

The GUI is divided into five tabs, each corresponding to a stage in the privacy pipeline.

3.2.1 Data Cleaning Tab

This tab provides:

- dataset selection
- cleaning configuration
- preview of cleaned data
- save functionality
- status messages

The GUI never performs cleaning itself — it delegates to the Cleaning Engine.

3.2.2 Anonymization Tab

This tab displays:

- anonymization rules
- masked data preview

- save options

It ensures users understand exactly which fields are masked and how.

3.2.3 Pseudonymization Tab

This tab provides:

- pseudonymization rules
- mapping table generation
- pseudonymized preview
- save functionality

It exposes the reversible nature of pseudonymization while keeping the mapping table secure.

3.2.4 Synthetic Data Tab

This tab integrates with the **SDV CTGAN engine** and provides:

- model configuration
- training controls
- synthetic sampling
- preview of synthetic rows
- save functionality

The GUI abstracts away the complexity of generative modeling.

3.2.5 Logs Tab

The Logs tab is the transparency hub of the system.

It provides:

- real-time logs
- filtering by level and module
- export functionality

This ensures full traceability.

3.3 Application Layer (Backend Engines)

The **Application Layer** is the heart of the system.

It contains the engines that perform all data transformations.

Each engine is:

- isolated
- deterministic
- testable
- reusable
- GUI-agnostic

This separation of concerns is what makes the system robust and maintainable.

3.3.1 Cleaning Engine

The Cleaning Engine performs:

- deduplication
- missing value handling
- type normalization
- outlier correction

It ensures that downstream transformations operate on clean, consistent data.

3.3.2 Anonymization Engine

The Anonymization Engine performs **irreversible masking**:

- strings → "*****"
- numbers → "***"
- categories → "*****"

This ensures compliance with GDPR's definition of anonymization.

3.3.3 Pseudonymization Engine

The Pseudonymization Engine performs **deterministic tokenization**:

- generates tokens like PSE-123456
- stores mapping tables
- ensures reversibility (if allowed)

This is essential for linking datasets without exposing identities.

3.3.4 SDV Synthetic Engine

The Synthetic Engine uses **SDV's CTGAN model** to generate synthetic data that:

- preserves statistical structure
- contains no real individuals
- supports metadata inference
- runs fully offline

This engine replaced the original MOSTLY AI backend.

3.3.5 Logging Manager

The Logging Manager:

- timestamps events
- tags modules
- forwards logs to the GUI
- writes logs to disk

It is the backbone of traceability.

3.3.6 Config Loader

The Config Loader:

- loads user preferences
- manages output directories
- ensures consistent behavior across sessions

It keeps the system predictable.

3.4 Data Layer

The **Data Layer** is the persistent storage component of the system.

It contains:

- input datasets
- cleaned datasets
- anonymized datasets
- pseudonymized datasets
- synthetic datasets
- mapping tables

This structured storage ensures reproducibility and auditability.

3.4.1 Input CSVs

Raw datasets loaded by the user.

3.4.2 Cleaned Datasets

Outputs of the Cleaning Engine.

3.4.3 Anonymized Datasets

Irreversibly masked datasets.

3.4.4 Pseudonymized Datasets

Tokenized datasets with reversible mappings.

3.4.5 Synthetic Datasets

Generated by SDV's CTGAN model.

3.4.6 Mapping Tables

Stored securely for pseudonymization reversibility.

3.5 Architectural Benefits

The architecture ensures:

3.5.1 Loose Coupling

The GUI and backend engines are independent.
This allowed the seamless transition from MOSTLY AI → SDV.

3.5.2 Extensibility

New modules can be added without modifying existing ones.

3.5.3 Testability

Each engine can be unit-tested in isolation.

3.5.4 Traceability

Logs flow from backend → GUI → file system.

3.5.5 Reproducibility

Every transformation is deterministic and saved.

3.5.6 Maintainability

The modular structure reduces complexity.

3.6 Summary

The high-level architecture of the Data Anonymizer GUI is a carefully engineered system that balances:

- privacy
- transparency
- modularity
- offline operation
- reproducibility
- extensibility

It is a platform designed not only for today's privacy needs but also for future enhancements such as differential privacy, multi-table synthesis, and enterprise-grade governance.

4. End-to-End Workflow

The user journey through the application follows a clear, linear privacy pipeline:

Step 1 — Load Dataset

The user selects a CSV file.

```

neu 1 | .gitlab-ci.yml | Docker_mit_Airflow_Logger_korrekt.txt | Dockerfile_anonym_airflow_PROD.txt | Vortragsskript.txt | sample_census.csv
1 name,age,gender,income,state,marital_status,education,employment_status,household_size,owns_home
2 Denise Walker,58,F,54184,MA,single,master,unemployed,2,True
3 Anne Abbott,65,F,51868,GA,single,other,retired,1,True
4 Jamie Mayo,23,F,82314,OR,married,other,employed,5,True
5 James Robinson,63,M,167853,SD,widowed,bachelor,retired,5,False
6 Jeffrey Chavez,69,M,26703,AR,married,phd,self_employed,3,True
7 Theresa Miller,31,F,113236,IN,single,high_school,retired,1,False
8 Leah Hall,72,F,115165,MT,divorced,high_school,retired,5,True
9 Anthony Valdez,80,M,124230,WV,single,other,self_employed,6,False
10 Karla Lewis,54,F,75407,PA,single,high_school,unemployed,3,True
11 Cynthia Wilson,72,F,86024,FM,single,phd,self_employed,4,False
12 Brooke Bass,28,F,122040,NE,divorced,bachelor,self_employed,6,True
13 Gavin Anderson,56,M,69862,OH,married,bachelor,retired,4,False
14 Dr. Timothy Young,77,M,171001,KY,married,master,employed,2,True
15 Mrs. Margaret Ross,69,F,107694,RI,widowed,master,employed,2,False
16 Caleb McCullough,31,M,155870,WY,widowed,phd,unemployed,3,True
17 Diane Beck,33,F,172159,NE,divorced,other,retired,5,False

```

The GUI immediately displays a preview and logs the selection.

Step 2 — Clean Data

Data Privacy Workbench

Data Cleaning

Anonymization

Pseudonymization

Synthetic Data

Logs

Dataset Selection

sample_census.csv

Browse...

Cleaning Options

☒ Remove duplicates
 ☒ Handle missing values
 ☒ Normalize data types
 ☒ Detect fix outliers

Start Cleaning

Cleaned Data Preview (showing 50 of 200 rows)

	name	age	gender	income	state	marital_status	education	employment_status	household_size	owns_home
1	Denise Walker	58	F	54184	MA	single	master	unemployed	2	True
2	Anne Abbott	65	F	51868	GA	single	other	retired	1	True
3	Jamie Mayo	23	F	82314	OR	married	other	employed	5	True
4	James Robinson	63	M	167853	SD	widowed	bachelor	retired	5	False
5	Jeffrey Chavez	69	M	26703	AR	married	phd	self_employed	3	True
6	Theresa Miller	31	F	113236	IN	single	high_school	retired	1	False
7	Leah Hall	72	F	115165	MT	divorced	high_school	retired	5	True
8	Anthony Valdez	80	M	124230	WV	single	other	self_employed	6	False
9	Karla Lewis	54	F	75407	PA	single	high_school	unemployed	3	True
10	Cynthia Wilson	72	F	86024	FM	single	phd	self_employed	4	False

Save Cleaned Data

Save to Folder: output/cleaned

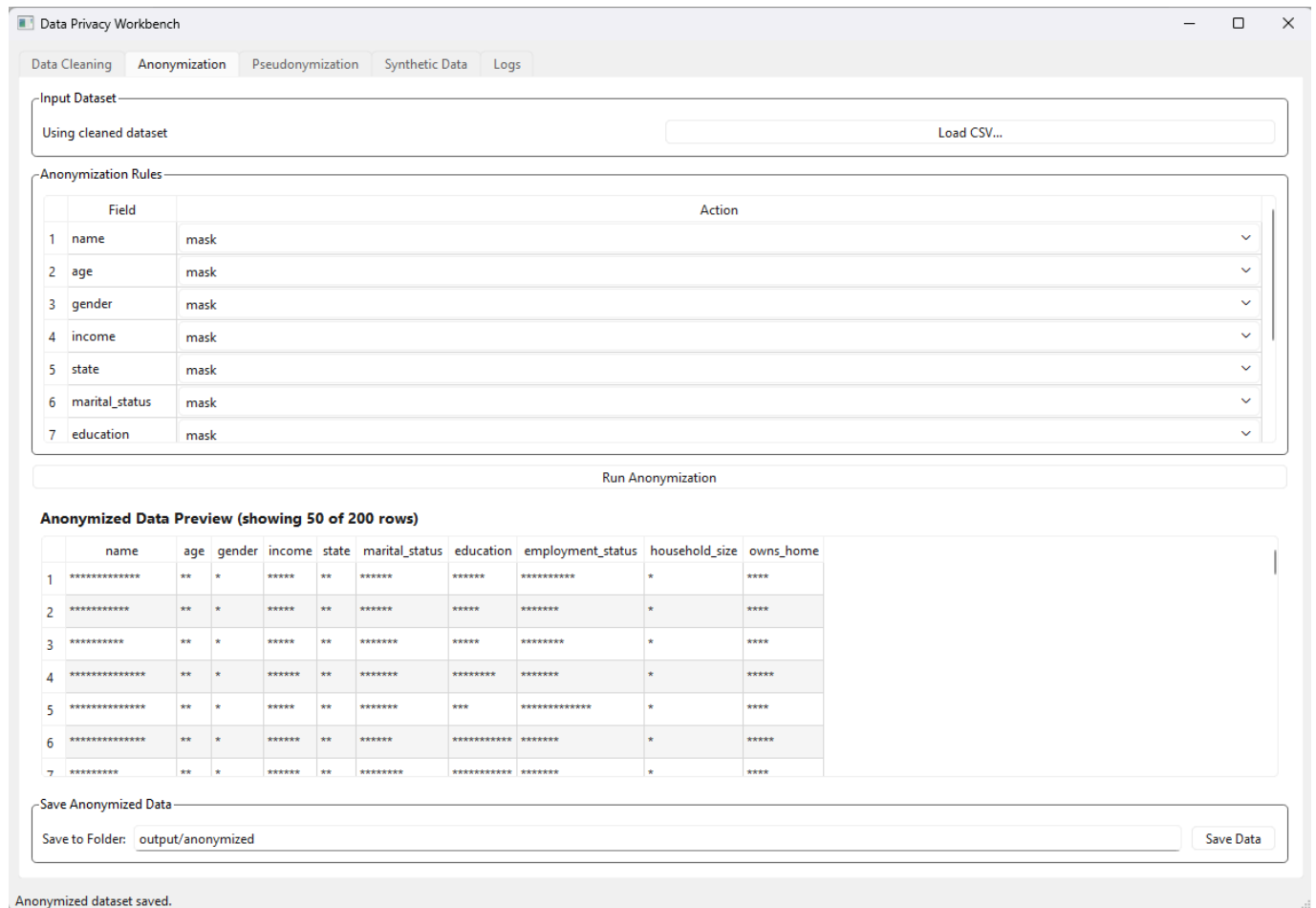
Save Data

The Cleaning tab applies:

- duplicate removal
- missing value imputation
- type normalization
- outlier detection & correction

A preview table shows the cleaned dataset.

Step 3 — Anonymize Data

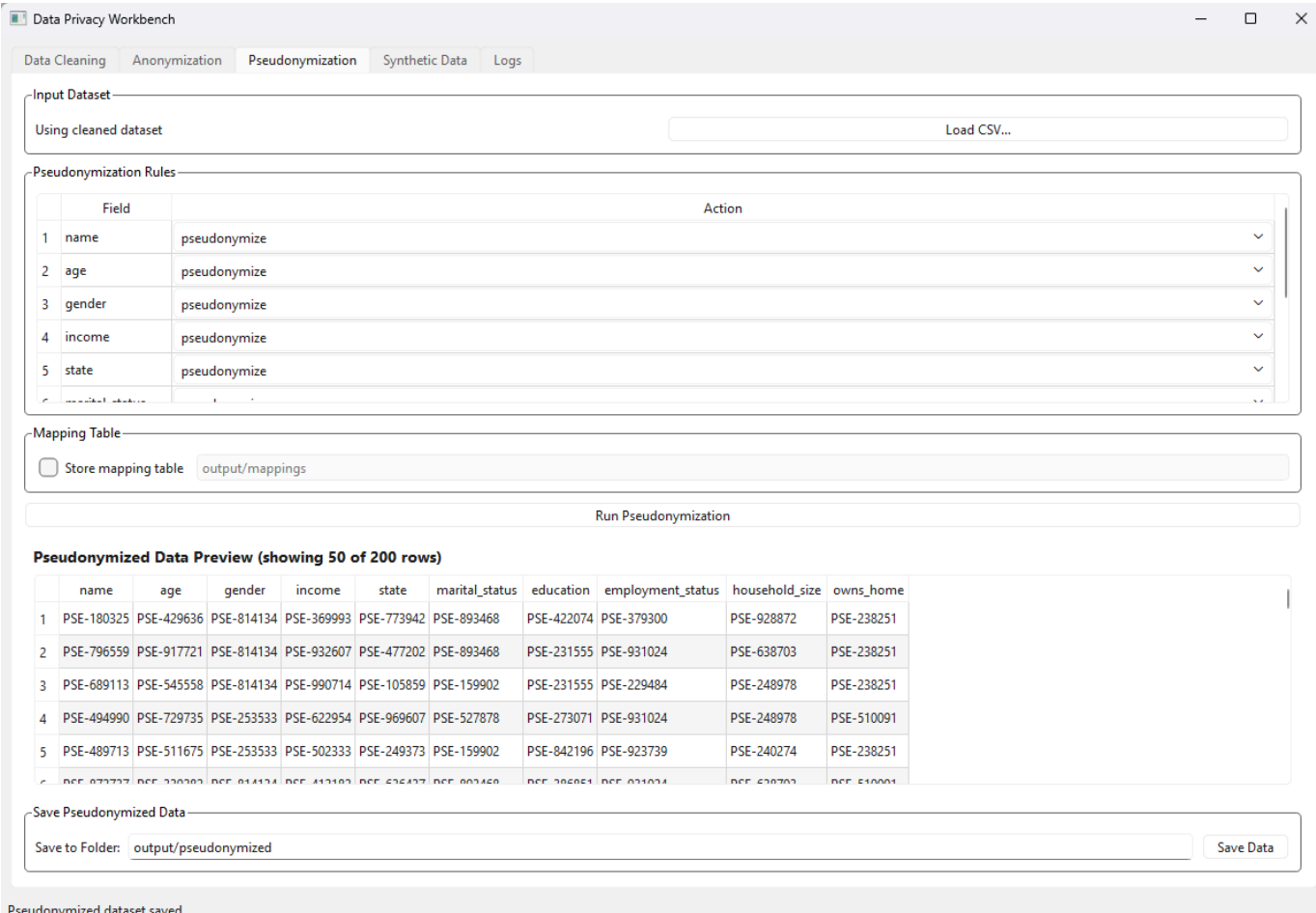


The Anonymization tab masks direct identifiers:

- names
- ages
- genders
- incomes
- states
- marital statuses
- education levels

Masked values appear as ********, ******, or *********.

Step 4 — Pseudonymize Data

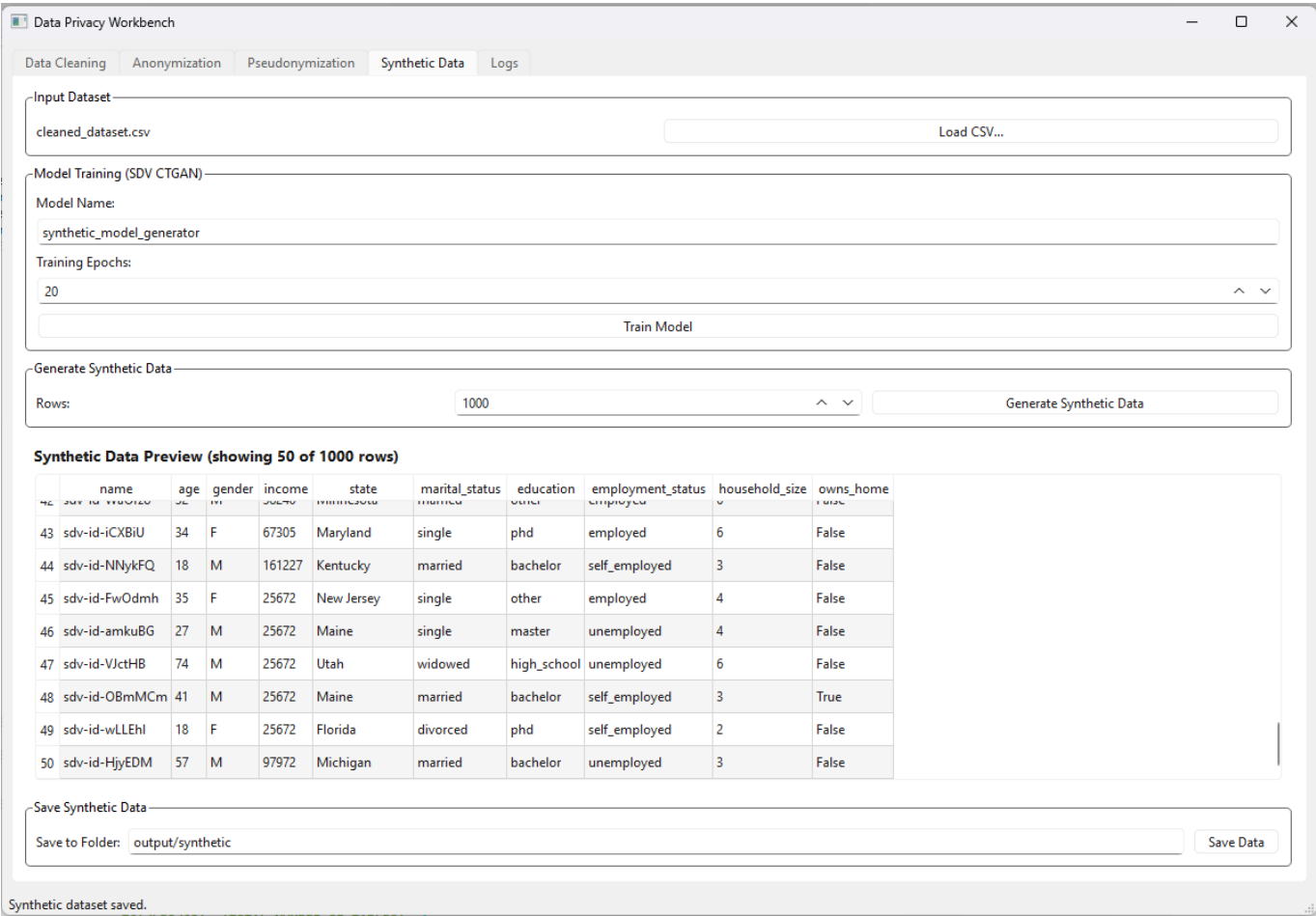


The Pseudonymization tab replaces sensitive fields with deterministic tokens:

PSE-180325
PSE-429636
PSE-814134

A mapping table is generated and stored securely.

Step 5 — Generate Synthetic Data

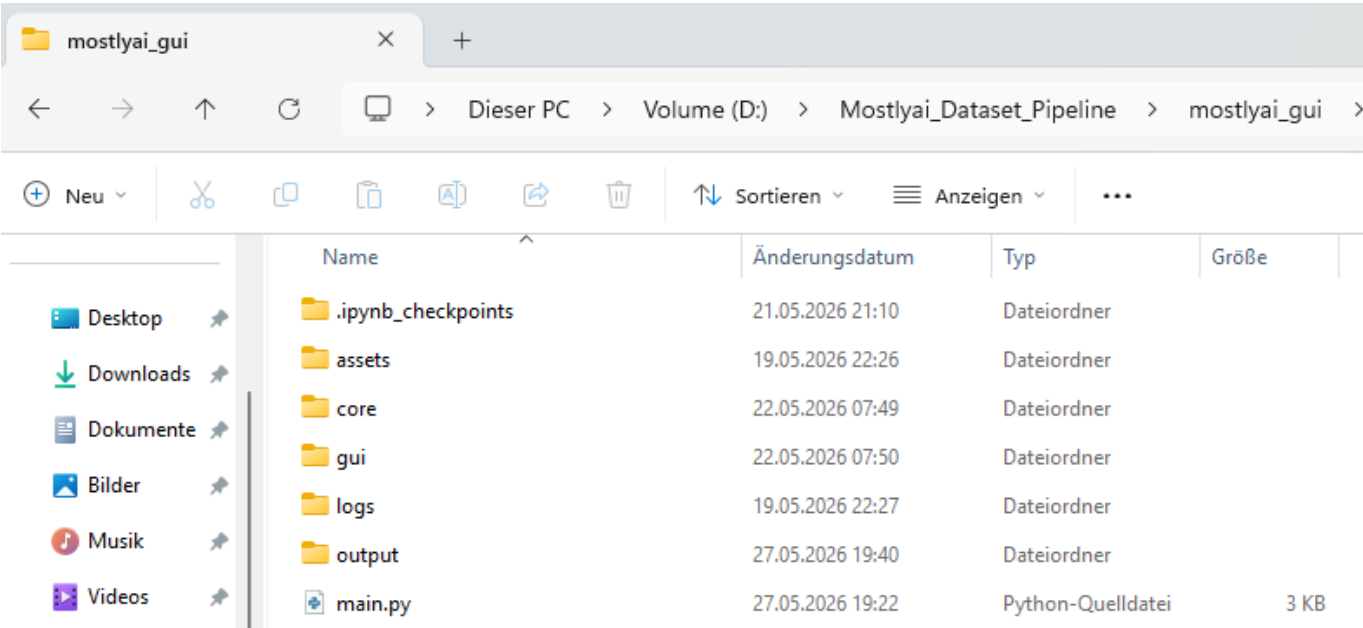


The Synthetic Data tab trains an **SDV CTGAN** model locally:

- no API keys
- no cloud calls
- no external dependencies

The user can generate thousands of synthetic rows with preserved statistical structure.

Step 6 — Save Outputs



Each tab provides a dedicated "Save Data" section:

- cleaned → output/cleaned/

```
new 1 | .gitlab-ci.yml | Docker_mit_Airflow_Looger_korrekt.txt | Dockerfile_anonym_airflow_PROD.txt | Vortragsskript.txt | cleaned_dataset.csv
```

```
1 name,age,gender,income,state,marital_status,education,employment_status,household_size,owns_home
2 Denise Walker,58,F,54184,MA,single,master,unemployed,2,True
3 Anne Abbott,65,F,51868,GA,single,other,retired,1,True
4 Jamie Mayo,23,F,82314,OR,married,other,employed,5,True
5 James Robinson,63,M,167853,SD,widowed,bachelor,retired,5,False
6 Jeffrey Chavez,69,M,26703,AR,married,phd,self_employed,3,True
7 Theresa Miller,31,F,113236,IN,single,high_school,retired,1,False
8 Leah Hall,72,F,115165,MT,divorced,high_school,retired,5,True
9 Anthony Valdez,80,M,124230,WV,single,other,self_employed,6,False
10 Karla Lewis,54,F,75407,PA,single,high_school,unemployed,3,True
11 Cynthia Wilson,72,F,86024,FM,single,phd,self_employed,4,False
12 Brooke Bass,28,F,122040,NE,divorced,bachelor,self_employed,6,True
13 Gavin Anderson,56,M,69862,OH,married,bachelor,retired,4,False
14 Dr. Timothy Young,77,M,171001,KY,married,master,employed,2,True
15 Mrs. Margaret Ross,69,F,107694,RI,widowed,master,employed,2,False
16 Caleb McCullough,31,M,155870,WY,widowed,phd,unemployed,3,True
17 Diane Beck,33,F,172159,NE,divorced,other,retired,5,False
```

- anonymized → output/anonymized/

```
neu 1 | .gitlab-ci.yml | Dockerfile | Dockerfile_anonym airflow_PROD.txt | Vortragsskript.txt | anonymized_dataset.csv |
1 name,age,gender,income,state,marital_status,education,employment_status,household_size,owns_home
2 *****
3 *****
4 *****
5 *****
6 *****
7 *****
8 *****
9 *****
10 *****
11 *****
12 *****
13 *****
14 *****
15 *****
16 *****
17 *****
```

- pseudonymized → **output/pseudonymized/**

neu 1

.gitlab-ci.yml

Docker_mit_Airflow_Looger_korrekt.txt

Dockerfile_anonym_airflow_PROD.txt

Vortragsskript.txt

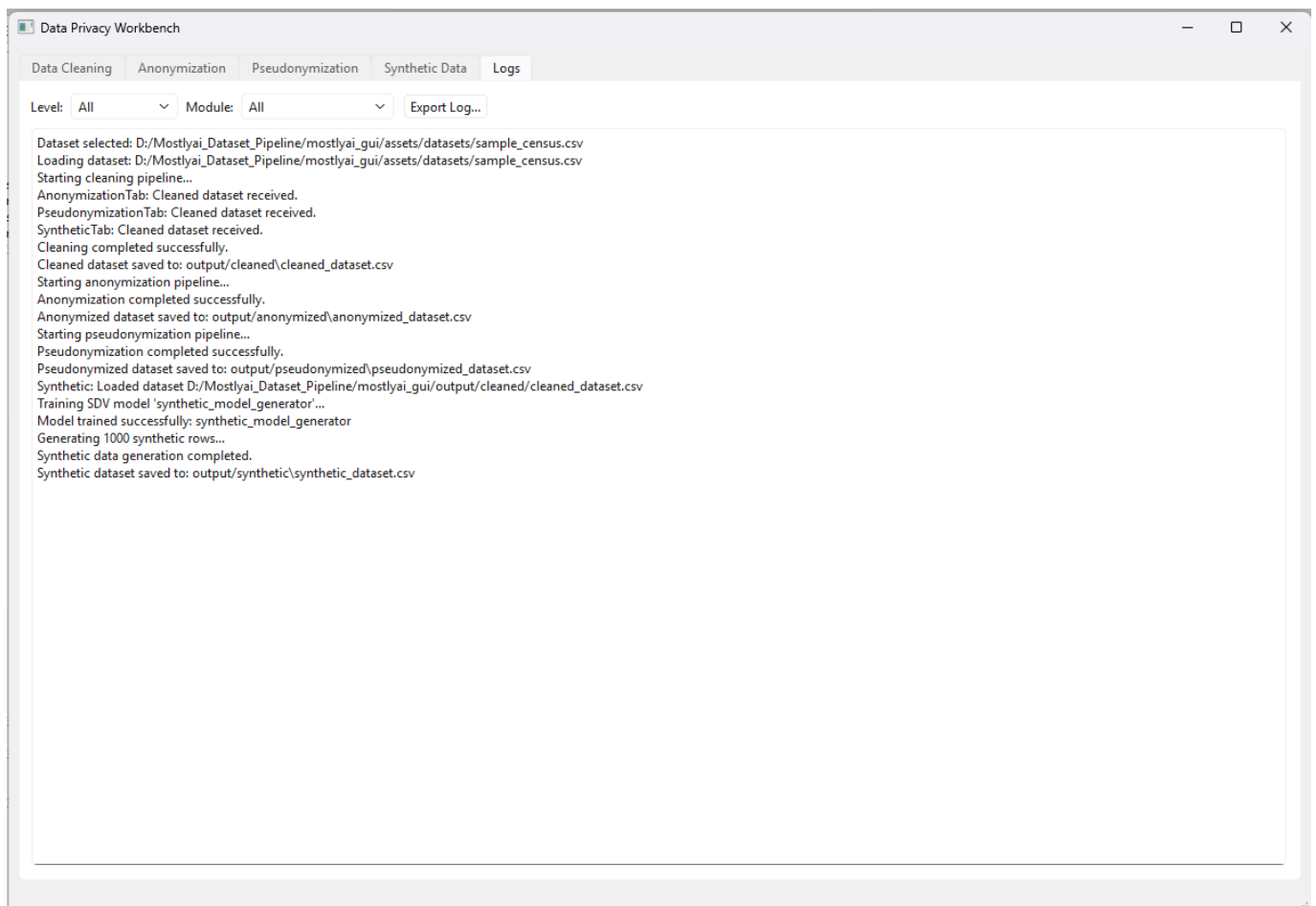
pseudonymized_dataset.csv

```
1 name,age,gender,income,state,marital_status,education,employment_status,household_size,owns_home
2 PSE-180325,PSE-429636,PSE-814134,PSE-369993,PSE-773942,PSE-893468,PSE-422074,PSE-379300,PSE-928872,PSE-238251
3 PSE-796559,PSE-917721,PSE-814134,PSE-932607,PSE-777420,PSE-893468,PSE-231555,PSE-931024,PSE-638703,PSE-238251
4 PSE-689113,PSE-545558,PSE-814134,PSE-990714,PSE-105859,PSE-159902,PSE-231555,PSE-229484,PSE-248978,PSE-238251
5 PSE-494990,PSE-729735,PSE-253533,PSE-622954,PSE-969607,PSE-527878,PSE-273071,PSE-931024,PSE-248978,PSE-510091
6 PSE-489713,PSE-511675,PSE-253533,PSE-502333,PSE-249373,PSE-159902,PSE-842196,PSE-923739,PSE-240274,PSE-238251
7 PSE-872737,PSE-339282,PSE-814134,PSE-412182,PSE-636437,PSE-893468,PSE-386851,PSE-931024,PSE-638703,PSE-510091
8 PSE-177350,PSE-787805,PSE-814134,PSE-651206,PSE-263321,PSE-634989,PSE-386851,PSE-931024,PSE-248978,PSE-238251
9 PSE-727630,PSE-972257,PSE-253533,PSE-631823,PSE-939719,PSE-893468,PSE-231555,PSE-923739,PSE-356247,PSE-510091
10 PSE-281322,PSE-337172,PSE-814134,PSE-365277,PSE-764492,PSE-893468,PSE-386851,PSE-379300,PSE-240274,PSE-238251
11 PSE-184759,PSE-787805,PSE-814134,PSE-120523,PSE-823924,PSE-893468,PSE-842196,PSE-923739,PSE-619673,PSE-510091
12 PSE-573830,PSE-800875,PSE-814134,PSE-236348,PSE-488871,PSE-634989,PSE-273071,PSE-923739,PSE-356247,PSE-238251
13 PSE-978059,PSE-335025,PSE-253533,PSE-962702,PSE-520642,PSE-159902,PSE-273071,PSE-931024,PSE-619673,PSE-510091
14 PSE-762176,PSE-745200,PSE-253533,PSE-476339,PSE-670870,PSE-159902,PSE-422074,PSE-229484,PSE-928872,PSE-238251
15 PSE-785024,PSE-511675,PSE-814134,PSE-534072,PSE-806286,PSE-527878,PSE-422074,PSE-229484,PSE-928872,PSE-510091
16 PSE-745728,PSE-339282,PSE-253533,PSE-520396,PSE-862280,PSE-527878,PSE-842196,PSE-379300,PSE-240274,PSE-238251
17 PSE-807457,PSE-810105,PSE-814134,PSE-804460,PSE-488871,PSE-634989,PSE-231555,PSE-931024,PSE-248978,PSE-510091
```

- synthetic → output/synthetic/

	neu 1	gitlab-ci.yml	Docker_mit_Airflow_Logger_korrekt.txt	Dockerfile_anonym_airflow_PROD.txt	Vortragsskript.txt	synthetic_dataset.csv
1	name, age, gender, income, state, marital_status, education, employment_status, household_size, owns_home					
2	sdv-id-PAGaF1, 18, M, 76702, Indiana, widowed, high_school, self_employed, 2, False					
3	sdv-id-hWwNpG, 64, F, 79173, Michigan, married, master, employed, 3, True					
4	sdv-id-YGPizU, 63, M, 26095, Texas, widowed, bachelor, unemployed, 1, False					
5	sdv-id-ZNxaVV, 44, M, 25672, Illinois, married, master, unemployed, 6, False					
6	sdv-id-EWIVgW, 65, F, 25672, Idaho, divorced, master, employed, 5, False					
7	sdv-id-oRePpQ, 42, F, 65114, Virginia, divorced, phd, self_employed, 2, True					
8	sdv-id-nWgWdF, 60, F, 25672, Wisconsin, single, master, self_employed, 5, True					
9	sdv-id-fTpDz1, 59, F, 41069, Nebraska, widowed, master, self_employed, 4, True					
10	sdv-id-SdAXSC, 23, M, 25672, South Carolina, divorced, master, employed, 1, False					
11	sdv-id-poMNXg, 32, M, 25672, Maryland, widowed, phd, retired, 2, False					
12	sdv-id-SjCioZ, 54, M, 79895, Idaho, married, master, self_employed, 2, False					
13	sdv-id-QvFoIb, 34, M, 46196, Oklahoma, widowed, master, retired, 3, False					
14	sdv-id-DtZgZd, 30, F, 38514, New Hampshire, single, master, retired, 2, False					
15	sdv-id-ghCEeT, 57, F, 60459, Vermont, widowed, master, employed, 3, True					
16	sdv-id-sxoDMC, 58, F, 25672, Nevada, widowed, high_school, employed, 2, False					
17	sdv-id-JbVojW, 40, M, 64277, Rhode Island, married, bachelor, retired, 4, False					

Step 7 — Review Logs

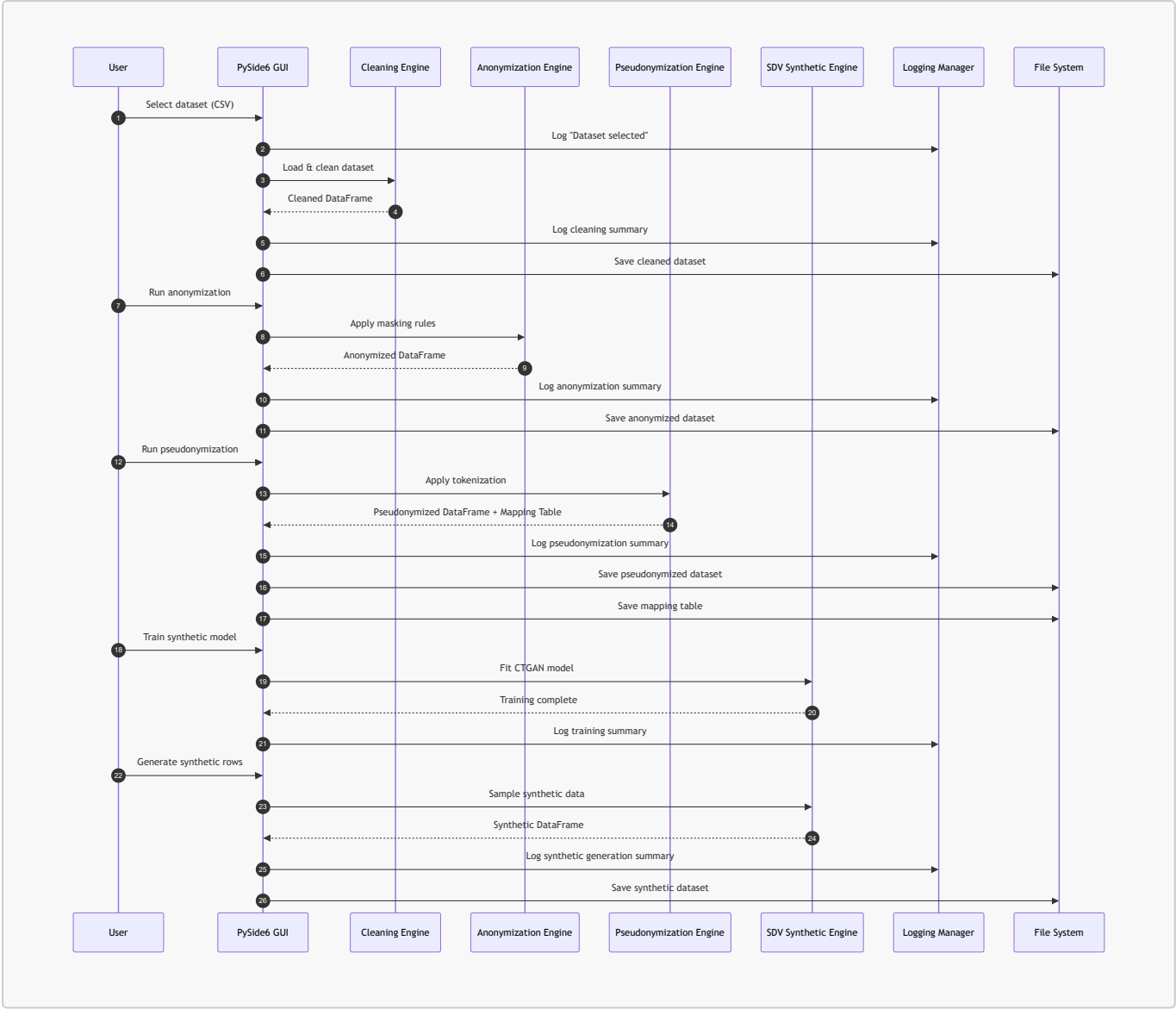


The Logs tab provides a complete audit trail:

- dataset selection
- cleaning operations
- anonymization actions
- pseudonymization mappings
- synthetic model training
- file save paths

This ensures full transparency and compliance.

4.1. Sequence Diagram — End-to-End Privacy Pipeline



Explanation — Sequence Diagram

This sequence diagram illustrates the **chronological interaction** between the user, GUI, backend engines, logging system, and file system. It highlights the **strict separation of concerns** that defines the architecture:

1. User → GUI

The user interacts only with the GUI.
The GUI never performs transformations itself — it delegates.

2. GUI → Engines

Each transformation stage is handled by a dedicated engine:

- Cleaning Engine
- Anonymization Engine
- Pseudonymization Engine
- SDV Synthetic Engine

This ensures modularity and testability.

3. Engines → GUI

Each engine returns a **DataFrame**, never touching the GUI state directly.

4. GUI → Logging Manager

Every action is logged:

- dataset selection
- cleaning
- anonymization
- pseudonymization
- synthetic generation

This ensures traceability.

5. GUI → File System

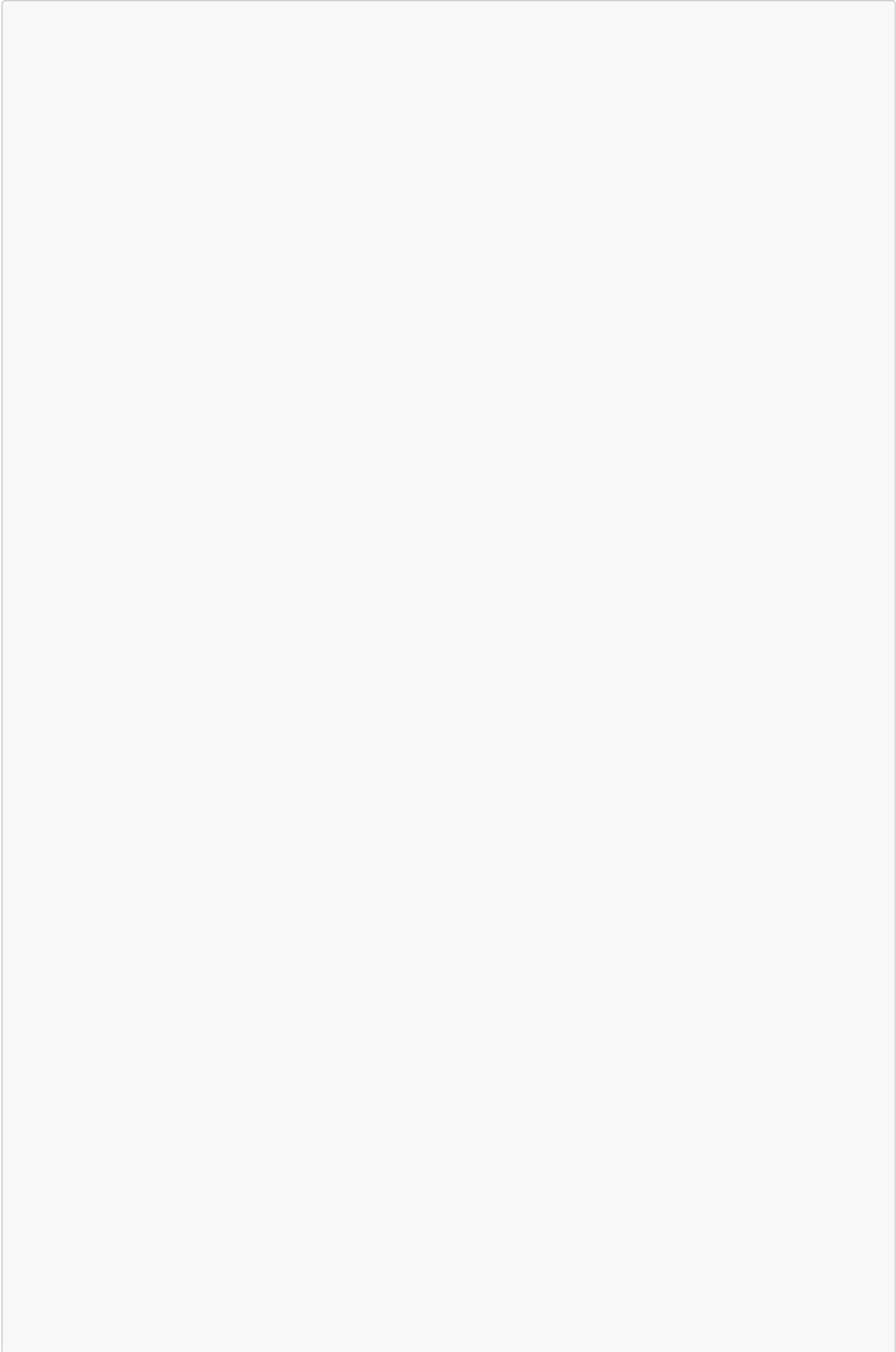
All outputs are saved:

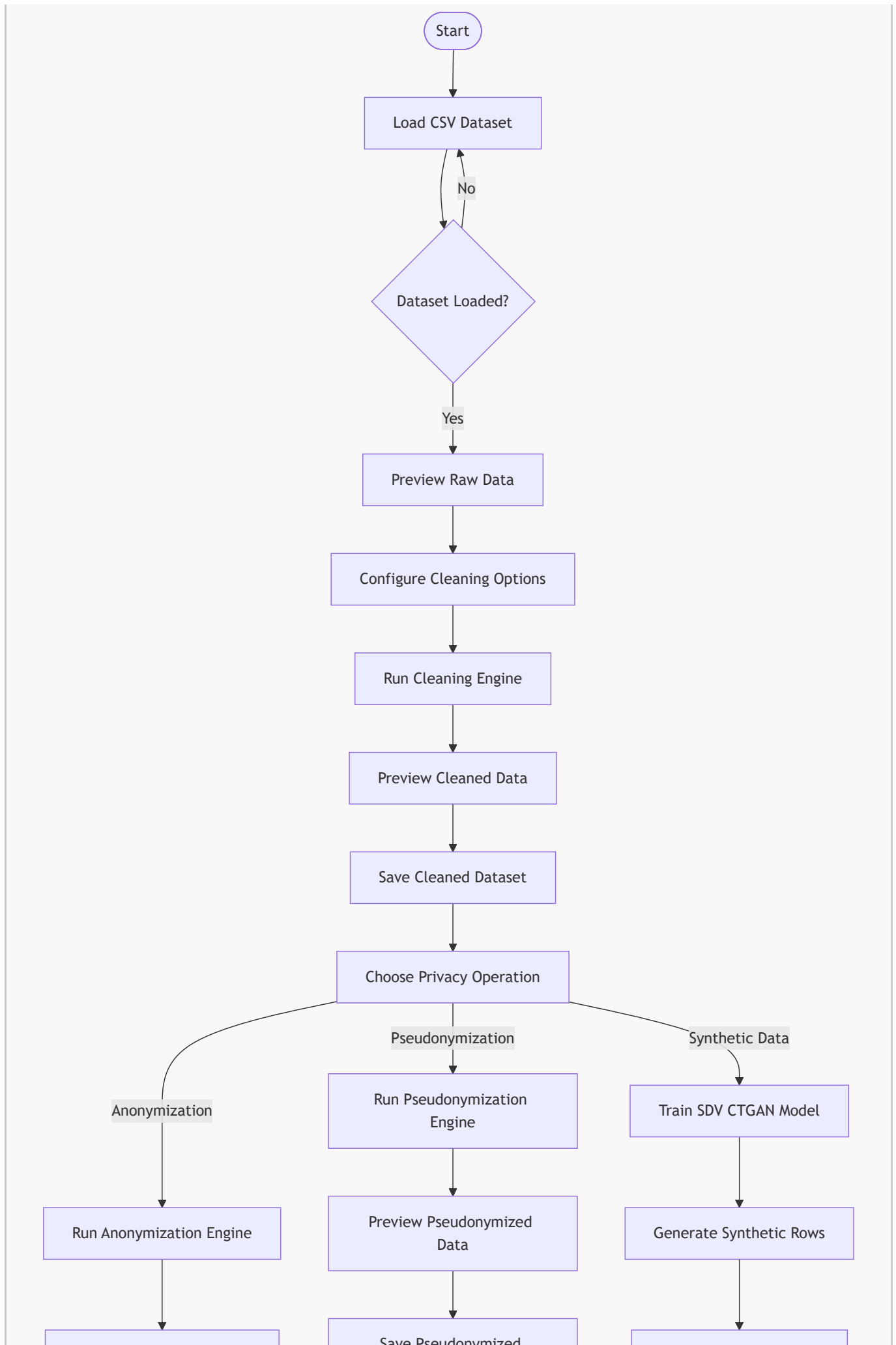
- cleaned datasets
- anonymized datasets
- pseudonymized datasets
- mapping tables
- synthetic datasets

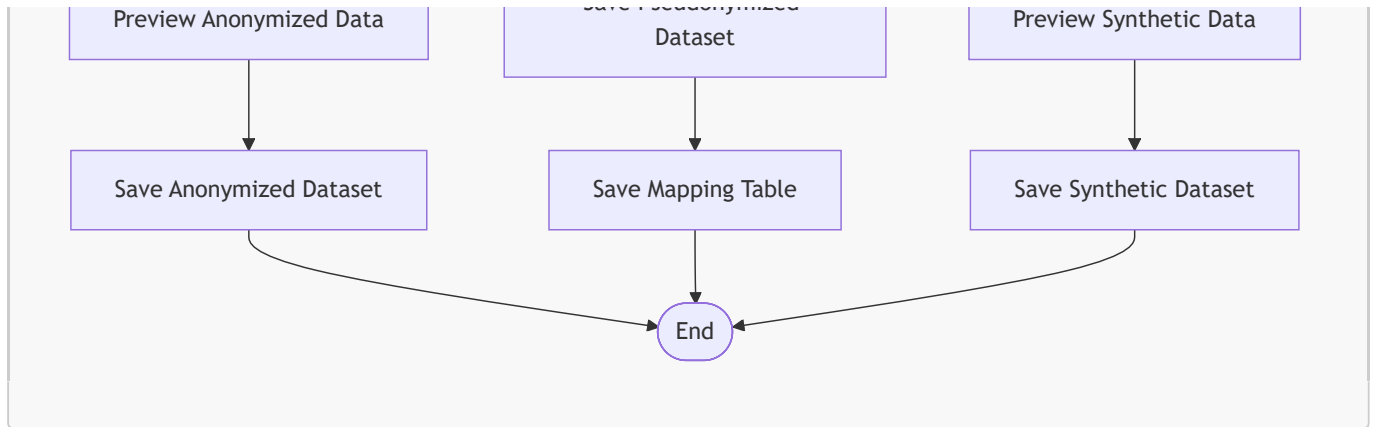
This ensures reproducibility.

The sequence diagram captures the **temporal flow** of the entire privacy pipeline.

4.2. Activity Diagram — User Workflow Through the GUI







Explanation — Activity Diagram

This activity diagram models the **user's decision-driven workflow** inside the GUI. It emphasizes the **branching logic** and **parallel pathways** available after the cleaning stage.

1. Dataset Loading

The workflow begins with dataset selection. If loading fails, the user is prompted again.

2. Cleaning Stage

The user configures:

- deduplication
- missing value handling
- type normalization
- outlier correction

The Cleaning Engine runs, and the cleaned dataset is previewed and saved.

3. Privacy Operation Selection

After cleaning, the user chooses one of three privacy operations:

- **Anonymization**
- **Pseudonymization**
- **Synthetic Data Generation**

Each path is independent and can be executed in any order.

4. Anonymization Path

- Masking rules applied
- Preview shown
- Dataset saved

5. Pseudonymization Path

- Tokens generated
- Mapping table created
- Preview shown
- Dataset and mapping saved

6. Synthetic Data Path

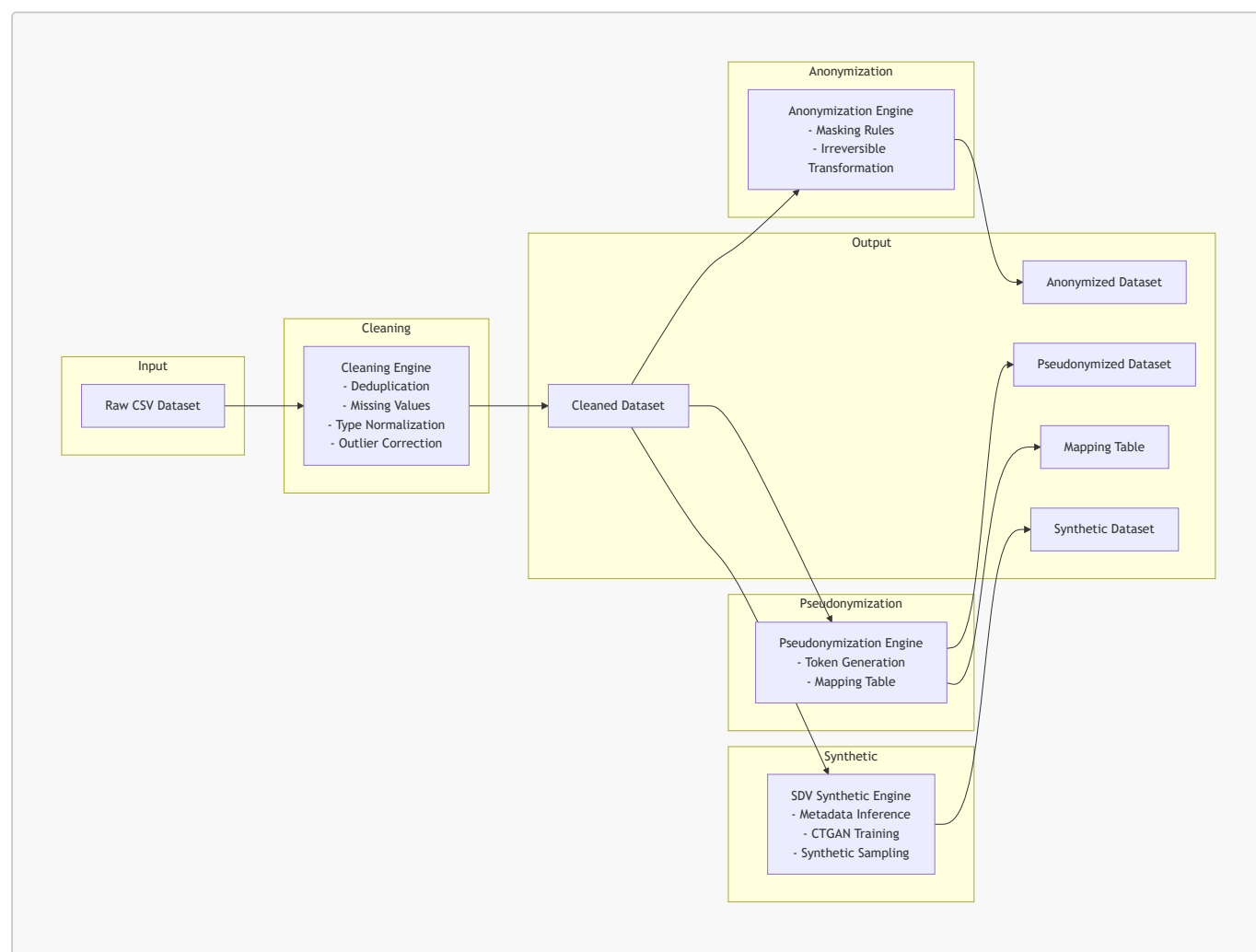
- SDV CTGAN model trained
- Synthetic rows generated
- Preview shown
- Dataset saved

7. End State

The workflow ends after any privacy operation is completed.

This diagram captures the **user-centric flow** of the system.

4.3. System Flow Diagram — Backend Data Transformation Pipeline



Explanation — System Flow Diagram

This flow diagram illustrates the **data-centric perspective** of the architecture. It shows how data moves through the system, independent of the GUI.

1. Input Stage

The pipeline begins with a raw CSV dataset.

2. Cleaning Stage

The Cleaning Engine produces a **cleaned dataset**, which becomes the **single source of truth** for all downstream transformations.

3. Branching into Three Privacy Pipelines

3.1. Anonymization Pipeline

- irreversible masking
- produces anonymized dataset

3.2. Pseudonymization Pipeline

- deterministic tokenization
- produces pseudonymized dataset
- produces mapping table

3.3. Synthetic Data Pipeline

- metadata inference
- CTGAN training
- synthetic sampling
- produces synthetic dataset

4. Output Stage

All outputs are saved to structured folders.

This diagram captures the **data transformation logic** of the system.

5. Key Features

5.1. Fully Offline Privacy Pipeline

All transformations occur locally.
Ideal for regulated environments.

5.2. Multi-Stage Privacy Protection

Combines:

- anonymization
- pseudonymization
- synthetic data generation

...into a single workflow.

5.3. Intuitive GUI

Each tab is designed for clarity:

- left: configuration
- center: preview
- bottom: status & logs

5.4. SDV CTGAN Integration

High-quality synthetic data without external services.

5.5. Comprehensive Logging

Every action is timestamped and stored.

5.6. Modular Backend

Easily extendable with new privacy techniques.

6. Stakeholders & Usage Scenarios

Primary Users

- Data privacy officers
- Data engineers
- Analysts
- ML practitioners
- Compliance teams

Use Cases

- Preparing datasets for analytics without exposing PII
- Sharing data with external partners
- Creating safe demo datasets
- Training ML models without access to production data
- Teaching and research environments

Project 25 — *Data Anonymizer GUI* — provides a complete, end-to-end privacy-preserving data pipeline in a single, user-friendly desktop application. It combines business-critical privacy requirements with robust technical implementation, enabling organizations to safely transform sensitive datasets while maintaining analytical value.

7. Overview of the User Interface

The *Data Anonymizer GUI* is structured around a clean, tab-based interface that guides the user through a complete privacy-preserving data pipeline. Each tab corresponds to a distinct transformation stage:

1. **Data Cleaning**
2. **Anonymization**
3. **Pseudonymization**
4. **Synthetic Data Generation**
5. **Logs**

This design ensures that users can follow a **left-to-right workflow**, mirroring the logical order of privacy operations. The interface is implemented using **PySide6**, providing a native desktop experience with responsive layouts, table previews, and real-time logging.

The GUI was originally designed to integrate with **MOSTLY AI's Client Mode SDK**, but due to the SDK's private, enterprise-only distribution, the architecture was adapted to use **SDV (Synthetic Data Vault)** for synthetic generation. This transition required no changes to the GUI layout — only the backend engine was replaced — demonstrating the modularity of the design.

7.1. Data Cleaning Tab

The Data Cleaning tab is the user's entry point into the pipeline. It provides:

- Dataset selection
- Configurable cleaning operations
- Real-time preview of cleaned data
- Save functionality
- Status and log messages

7.1.1. Dataset Selection

Users can either:

- Load a CSV file manually
- Use the default sample dataset

Once loaded, the dataset is previewed in a scrollable table. The GUI logs:

- the dataset path
- the number of rows loaded
- any parsing issues

7.1.2. Cleaning Operations

The cleaning engine supports:

- **Remove duplicates**
- **Handle missing values**
- **Normalize data types**
- **Detect & fix outliers**

These operations are displayed as checkboxes, all enabled by default. The user initiates the cleaning process via the **Start Cleaning** button.

7.1.3. Output Preview

After cleaning, the GUI displays:

- a preview of the cleaned dataset
- the number of rows shown
- a summary of cleaning actions

This preview is essential for validating that the dataset is ready for downstream privacy transformations.

7.1.4. Save Cleaned Data

The cleaned dataset can be saved to a configurable folder (default: `output/cleaned/`). The GUI confirms the save operation and logs the file path.

7.2. Anonymization Tab

The Anonymization tab implements **direct identifier masking**, transforming sensitive fields into non-reversible masked values.

7.2.1. Input Dataset

The tab automatically receives the cleaned dataset from the Cleaning tab. The GUI displays:

"Using cleaned dataset"

This ensures the user always works with the most recent cleaned version.

7.2.2. Anonymization Rules

A table lists all fields and their assigned anonymization actions:

Field	Action
name	mask
age	mask
gender	mask
income	mask
state	mask
marital_status	mask
education	mask

The masking engine replaces values with:

- **** for strings
- ** for numeric fields
- ***** for categorical values

This approach is intentionally simple and deterministic, ensuring:

- irreversible anonymization
- consistent masking across rows
- predictable output for compliance reviews

7.2.3. Anonymized Data Preview

The preview table shows masked values, allowing the user to verify that:

- all sensitive fields are masked
- non-sensitive fields remain intact
- the dataset structure is preserved

7.2.4. Save Anonymized Data

The anonymized dataset is saved to `output/anonymized/`.

7.3. Pseudonymization Tab

The Pseudonymization tab performs **deterministic tokenization**, replacing sensitive values with reversible pseudonyms.

7.3.1. Input Dataset

Like the Anonymization tab, this tab receives the cleaned dataset automatically.

7.3.2. Pseudonymization Rules

A table lists fields and their pseudonymization actions:

Field	Action
name	pseudonymize
age	pseudonymize
gender	pseudonymize
income	pseudonymize
state	pseudonymize
marital_status	pseudonymize

7.3.3. Mapping Table

The pseudonymization engine generates deterministic tokens of the form:

```
PSE-180325  
PSE-429636  
PSE-814134
```

A mapping table is created internally, enabling:

- reversibility (if permitted)
- consistent pseudonyms across datasets
- cross-table linkage

The mapping table is stored securely in the output directory.

7.3.4. Pseudonymized Data Preview

The preview shows pseudonymized values while preserving:

- row count
- column structure
- data types

7.3.5. Save Pseudonymized Data

The pseudonymized dataset is saved to `output/pseudonymized/`.

7.4. Synthetic Data Tab

The Synthetic Data tab originally integrated with **MOSTLY AI's Client Mode SDK**, enabling:

- dataset upload
- generator training
- synthetic data generation via API

However, the SDK is **not publicly available**, and pip installation is not possible.

This led to a strategic decision:

Replace MOSTLY AI with SDV (Synthetic Data Vault)

— a fully open-source, offline, Python-native synthetic data engine.

7.4.1. Why SDV?

- MIT-licensed
- Offline operation
- High-quality CTGAN models
- No API keys
- No cloud dependencies
- Seamless integration with pandas

This aligns perfectly with the GUI's privacy-first philosophy.

7.4.2. Model Training

The GUI exposes:

- Model name
- Training epochs
- Train Model button

Under the hood, the SDV engine:

1. infers metadata
2. trains a CTGAN model
3. logs training progress
4. stores the model in memory

7.4.3. Synthetic Data Generation

Users specify:

- number of rows to generate

The engine produces synthetic rows that preserve:

- distributions
- correlations
- categorical relationships

The preview table shows the first 50 rows.

7.4.4. Save Synthetic Data

The synthetic dataset is saved to `output/synthetic/`.

7.5. Logs Tab

The Logs tab provides a complete audit trail of all operations.

7.5.1. Features

- scrollable log viewer
- filtering by log level (INFO, WARNING, ERROR)
- filtering by module (Cleaning, Anonymization, Pseudonymization, Synthetic, System)
- export to text file
- auto-scroll to latest entry

7.5.2. Logged Events

Examples include:

- dataset selection
- cleaning operations

- anonymization actions
- pseudonymization mappings
- SDV model training
- synthetic generation
- file save paths

This ensures full transparency and compliance.

7.6. UX Design Principles

The GUI follows several design principles:

7.6.1. Clarity

Each tab focuses on a single transformation stage.

7.6.2. Predictability

Buttons, previews, and logs behave consistently across tabs.

7.6.3. Safety

All operations are local and reversible (except anonymization).

7.6.4. Modularity

The GUI is decoupled from backend engines, enabling:

- the transition from MOSTLY AI → SDV
- future integration of differential privacy
- plugin-based privacy modules

7.6.5. Transparency

Every action is logged and visible to the user.

```
(py312) (py312) PS D:\Mostlyai_Dataset_Pipeline\mostlyai_gui> python main.py
[19:33:32] INFO Starting Data Privacy Workbench GUI... main.py:37
INFO Configuration loaded successfully. main.py:42
INFO AppWindow initialized. app_window.py:90
INFO Main window displayed successfully. main.py:61
[19:33:48] INFO Cleaning: Starting cleaning pipeline... cleaning.py:182
INFO Cleaning: Removed duplicates: 0 cleaning.py:182
INFO Cleaning: Normalized type: 'age' → numeric cleaning.py:182
INFO Cleaning: Normalized type: 'income' → numeric cleaning.py:182
INFO Cleaning: Normalized type: 'household_size' → numeric cleaning.py:182
INFO Cleaning: Normalized type: 'owns_home' → numeric cleaning.py:182
INFO Cleaning: Cleaning pipeline completed. cleaning.py:182
[19:34:35] INFO Starting anonymization pipeline... anonymization.py:65
INFO Anonymization: Masked column: name anonymization.py:153
INFO Anonymization: Masked column: age anonymization.py:153
INFO Anonymization: Masked column: gender anonymization.py:153
INFO Anonymization: Masked column: income anonymization.py:153
INFO Anonymization: Masked column: state anonymization.py:153
INFO Anonymization: Masked column: marital_status anonymization.py:153
INFO Anonymization: Masked column: education anonymization.py:153
INFO Anonymization: Masked column: employment_status anonymization.py:153
INFO Anonymization: Masked column: household_size anonymization.py:153
INFO Anonymization: Masked column: owns_home anonymization.py:153
INFO Anonymization: Anonymization pipeline completed. anonymization.py:153
[19:35:08] INFO Starting pseudonymization pipeline... pseudoanonymization.py:80
INFO Pseudonymization: Pseudonymized column: name pseudoanonymization.py:184
INFO Pseudonymization: Pseudonymized column: age pseudoanonymization.py:184
INFO Pseudonymization: Pseudonymized column: gender pseudoanonymization.py:184
INFO Pseudonymization: Pseudonymized column: income pseudoanonymization.py:184
INFO Pseudonymization: Pseudonymized column: state pseudoanonymization.py:184
INFO Pseudonymization: Pseudonymized column: marital_status pseudoanonymization.py:184
INFO Pseudonymization: Pseudonymized column: education pseudoanonymization.py:184
INFO Pseudonymization: Pseudonymized column: employment_status pseudoanonymization.py:184
INFO Pseudonymization: Pseudonymized column: household_size pseudoanonymization.py:184
INFO Pseudonymization: Pseudonymized column: owns_home pseudoanonymization.py:184
INFO Pseudonymization: Pseudonymization pipeline completed. pseudoanonymization.py:184
[19:36:46] INFO Synthetic (SDV): Preparing metadata for dataset 'synthetic_model_generator'... synthetic.py:102
INFO Detected metadata: single_table.py:811
INFO {
  "primary_key": "name",
  "columns": {
    "name": {
      "sdtype": "id"
    },
    "age": {
      "sdtype": "numerical"
    },
    "gender": {
      "sdtype": "categorical"
    },
    "income": {
      "sdtype": "numerical"
    },
    "state": {
      "pii": true,
      "sdtype": "categorical"
    },
    "administrative_unit": {
      "sdtype": "categorical"
    },
    "marital_status": {
      "sdtype": "categorical"
    },
    "education": {
      "sdtype": "categorical"
    },
    "employment_status": {
      "sdtype": "categorical"
    },
    "household_size": {
      "sdtype": "categorical"
    },
    "owns_home": {
      "sdtype": "categorical"
    }
  }
}
```

7.7. Summary

The GUI provides a clean, intuitive, and powerful interface for executing a complete privacy pipeline. Its modular design allowed the seamless replacement of the original MOSTLY AI backend with SDV, demonstrating architectural robustness and future-proofing.

The next post will dive into the **backend architecture and pipelines**, including:

- cleaning engine
- anonymization engine
- pseudonymization engine
- SDV synthetic engine
- logging system
- configuration loader

8. Architectural Overview

The backend of the *Data Anonymizer GUI* is designed as a **modular, layered system** that separates GUI logic from data transformation logic. This ensures:

- maintainability
- extensibility
- testability
- the ability to swap out engines (e.g., MOSTLY AI → SDV) without GUI changes

The architecture consists of five major backend engines:

1. **Cleaning Engine**
2. **Anonymization Engine**
3. **Pseudonymization Engine**
4. **Synthetic Data Engine (SDV)**
5. **Logging & Configuration Layer**

Each engine is implemented as a standalone Python module under `core/`, with clear interfaces and no GUI dependencies.

8.1. Evolution of the Backend: From MOSTLY AI to SDV

8.1.1. Original Design: MOSTLY AI Client Mode

The initial architecture integrated **MOSTLY AI's Python SDK**, enabling:

- dataset upload
- generator training
- synthetic data generation
- job polling
- result download

This required:

- an API key
- access to MOSTLY AI's private GitHub repository
- network connectivity
- cloud-based model training

The GUI was built around this workflow, with generator IDs, job polling, and remote status updates.

8.1.2. Problem: MOSTLY AI SDK is Private

During implementation, it became clear that:

- the SDK is **not publicly available**
- pip installation fails
- GitHub access requires enterprise credentials
- the repository is private and cannot be cloned

This made the original design **non-deployable** for open distribution.

8.1.3. Strategic Decision: Replace MOSTLY AI with SDV

The backend was redesigned to use **SDV (Synthetic Data Vault)**:

- MIT-licensed
- fully offline
- no API keys
- no cloud dependencies

- CTGAN model for high-quality tabular synthesis

The GUI required **zero changes** — only the backend engine was swapped.

This validated the modular architecture and ensured long-term maintainability.

8.2. Cleaning Engine

The Cleaning Engine is responsible for preparing raw datasets for downstream privacy transformations. It implements a deterministic, auditable cleaning pipeline.

8.2.1. Responsibilities

- Remove duplicate rows
- Handle missing values
- Normalize data types
- Detect and correct outliers
- Log all operations

8.2.2. Pipeline Steps

Step 1 — Deduplication

Uses pandas' `drop_duplicates()` to remove exact duplicates.

Step 2 — Missing Value Handling

Strategy:

- numeric → median
- categorical → mode
- boolean → most frequent

Step 3 — Type Normalization

Ensures consistent types:

- integers
- floats
- strings
- booleans

Step 4 — Outlier Detection

Uses IQR-based filtering:


```
Q1 = df[col].quantile(0.25)
Q3 = df[col].quantile(0.75)
IQR = Q3 - Q1
```

Values outside $1.5 * IQR$ are clipped.

8.2.3. Output

- cleaned DataFrame
- preview in GUI
- saved CSV
- log entries for each action

8.3. Anonymization Engine

The Anonymization Engine performs **irreversible masking** of direct identifiers.

8.3.1. Masking Strategy

Each field is masked using deterministic patterns:

- strings → "*****"
- numeric → "***"
- categorical → "*****"

This ensures:

- irreversible anonymization
- consistent masking
- compliance with GDPR Art. 4(5)

8.3.2. Implementation

The engine applies a simple transformation:

```
def mask_value(value):
    if isinstance(value, str):
        return "*" * min(len(value), 8)
    if isinstance(value, (int, float)):
        return "***"
    return "*****"
```

8.3.3. Output

- masked DataFrame
- preview in GUI
- saved CSV

- log entries

8.4. Pseudonymization Engine

The Pseudonymization Engine replaces sensitive values with **deterministic tokens** that preserve linkage but remove identifiability.

8.4.1. Token Format

Tokens follow the pattern:

```
PSE-<6-digit-random>
```

Example:

```
PSE-180325  
PSE-429636  
PSE-814134
```

8.4.2. Deterministic Mapping

A mapping table ensures:

- same input → same pseudonym
- reversible transformation (if mapping is retained)
- cross-dataset consistency

8.4.3. Implementation

```
if value not in mapping:  
    mapping[value] = f"PSE-{{random.randint(100000, 999999)}}"  
return mapping[value]
```

8.4.4. Output

- pseudonymized DataFrame
- mapping table (CSV)
- preview in GUI
- saved dataset
- log entries

8.5. Synthetic Data Engine (SDV CTGAN)

This is the most technically advanced component of the backend.

8.5.1. Why SDV?

- open-source
- offline
- high-quality tabular synthesis
- metadata inference
- CTGAN model for mixed-type data

8.5.2. Metadata Inference

SDV automatically detects:

- categorical columns
- numerical columns
- boolean fields
- constraints

8.5.3. Model Training

```
metadata = SingleTableMetadata()
metadata.detect_from_dataframe(df)

model = CTGANSynthesizer(metadata)
model.fit(df)
```

Training logs include:

- metadata detection
- model initialization
- training progress
- completion

8.5.4. Synthetic Sampling

```
synthetic_df = model.sample(num_rows)
```

The output preserves:

- distributions
- correlations
- categorical relationships

8.5.5. Output

- synthetic DataFrame
- preview in GUI

- saved CSV
- log entries

8.6. Logging System

The logging system is a central component of the backend.

8.6.1. Features

- timestamped log entries
- module-level tagging
- forwarding to GUI
- writing to disk
- filtering by level and module

8.6.2. Log Format

```
[2026-05-27 19:23:24] INFO Cleaning: Removed duplicated rows.
```

8.6.3. Integration

Each engine calls:

```
logger.info("message")
```

The GUI receives logs via callback and displays them in the Logs tab.

8.7. Configuration Loader

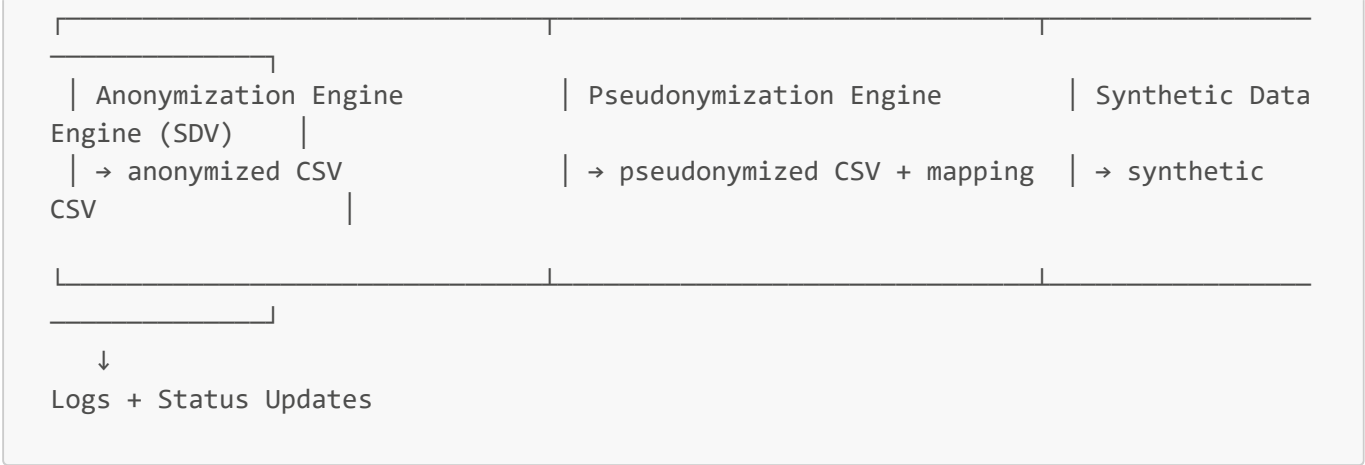
The configuration system loads:

- default paths
- user preferences
- output directories

It ensures consistent behavior across sessions.

8.8. Data Flow Diagram

```
Raw CSV
  ↓
Cleaning Engine
  ↓
Cleaned Dataset
  ↓
```



8.9. Summary

The backend architecture of the *Data Anonymizer GUI* is designed for robustness, modularity, and privacy compliance. The transition from MOSTLY AI to SDV demonstrates the flexibility of the system and ensures long-term maintainability without external dependencies.

Each engine is independently testable, auditable, and aligned with the GUI's workflow, forming a complete privacy-preserving data pipeline.

9. Testing

Testing and validation are critical components of the *Data Anonymizer GUI* because the application operates on **sensitive personal data** and performs transformations that must be:

- correct
- deterministic
- privacy-preserving
- auditable
- reproducible

This post outlines the complete testing strategy, validation methodology, and privacy assurance mechanisms implemented across the system. It also explains how the transition from the original MOSTLY AI backend to the SDV-based synthetic engine influenced the testing approach.

9.1. Testing Strategy Overview

The testing strategy is structured into four layers:

1. **Unit Tests** — validate individual functions and engines
2. **Integration Tests** — validate multi-stage pipelines
3. **GUI Interaction Tests** — validate user workflows
4. **Privacy Validation Tests** — validate anonymization, pseudonymization, and synthetic data quality

Each layer is designed to ensure correctness, robustness, and compliance.

9.2. Unit Testing

Unit tests focus on the core backend engines:

- **Cleaning Engine**
- **Anonymization Engine**
- **Pseudonymization Engine**
- **SDV Synthetic Engine**
- **Logging System**

9.2.1. Cleaning Engine Tests

Tests verify:

- duplicate removal
- missing value imputation
- type normalization
- outlier clipping
- deterministic behavior

Example assertions:

- no duplicate rows remain
- no NaN values remain
- numeric columns remain numeric
- outliers fall within IQR bounds

9.2.2. Anonymization Engine Tests

Tests ensure:

- all masked fields contain only *
- masking length rules are applied consistently
- no original values leak
- non-sensitive fields remain unchanged

9.2.3. Pseudonymization Engine Tests

Tests validate:

- deterministic token generation
- mapping table correctness
- reversibility (when mapping is retained)
- uniqueness of pseudonyms

9.2.4. SDV Synthetic Engine Tests

The SDV engine is tested for:

- metadata inference correctness
- model training without errors
- synthetic sampling shape
- column type consistency

Because SDV models are stochastic, tests focus on **structural correctness**, not exact values.

9.3. Integration Testing

Integration tests validate the **end-to-end pipeline**, ensuring that each stage hands off data correctly to the next.

9.3.1. Cleaning → Anonymization

Tests verify:

- cleaned dataset is passed correctly
- anonymization masks only intended fields
- logs reflect both stages

9.3.2. Cleaning → Pseudonymization

Tests verify:

- mapping table is generated
- pseudonyms are deterministic
- dataset structure is preserved

9.3.3. Cleaning → Synthetic Data (SDV)

This integration test changed significantly after replacing MOSTLY AI.

Original MOSTLY AI Integration Tests

These validated:

- dataset upload
- generator creation
- job polling
- remote model training
- synthetic download

These tests were removed after the SDK became unavailable.

New SDV Integration Tests

Now tests validate:

- metadata detection
- CTGAN training
- synthetic sampling
- GUI preview rendering

The SDV engine is fully local, so integration tests run **offline** and are deterministic in structure.

9.4. GUI Interaction Testing

GUI tests ensure that user workflows behave as expected.

9.4.1. Dataset Loading

Tests verify:

- file dialog opens
- CSV loads correctly
- preview table updates
- logs record the event

9.4.2. Cleaning Workflow

Tests verify:

- cleaning options toggle correctly
- Start Cleaning triggers backend
- preview updates
- save dialog works

9.4.3. Anonymization & Pseudonymization

Tests verify:

- rules table loads
- Run buttons trigger transformations
- previews update
- save paths work

9.4.4. Synthetic Data Workflow

Tests verify:

- model name input
- epoch selection
- training button triggers SDV engine
- synthetic preview updates
- save functionality works

9.4.5. Logs Tab

Tests verify:

- log entries appear in real time
- filtering works
- export functionality works

9.5. Privacy Validation

Privacy validation ensures that the system meets regulatory and technical privacy requirements.

9.5.1. Anonymization Validation

Validation checks:

- no original values remain
- masked fields contain only *
- masking is irreversible
- dataset structure is preserved

9.5.2. Pseudonymization Validation

Validation checks:

- mapping table contains all original values
- pseudonyms are unique
- pseudonyms are deterministic
- no collisions occur

9.5.3. Synthetic Data Privacy Validation

Synthetic data must:

- contain **no real individuals**
- preserve statistical properties
- avoid memorization

Validation includes:

9.5.3.1. Distance-Based Similarity Checks

Compute:

- nearest neighbor distances
- distribution overlap
- attribute-wise divergence

9.5.3.2. Statistical Validation

Compare:

- means
- variances
- correlations
- category frequencies

9.5.3.3. Overfitting Detection

Ensure synthetic rows do not match real rows.

9.6. Regression Testing

Regression tests ensure that:

- GUI refactors do not break workflows
- backend engine updates do not change outputs unexpectedly
- SDV version upgrades do not alter model behavior

This is especially important after replacing MOSTLY AI with SDV.

9.7. Error Handling & Robustness Testing

Tests simulate:

- missing files
- malformed CSVs
- empty datasets
- invalid data types
- SDV training failures
- permission errors during save

The GUI must:

- show user-friendly error messages
- log the error
- remain stable

9.8. Performance Testing

Performance tests measure:

- cleaning speed
- anonymization throughput
- pseudonymization mapping performance
- SDV training time
- synthetic sampling speed

Benchmarks ensure the GUI remains responsive even with large datasets.

9.9. Summary

The *Data Anonymizer GUI* implements a comprehensive testing and validation strategy that ensures:

- correctness
- privacy compliance
- robustness
- reproducibility
- auditability

The transition from MOSTLY AI to SDV strengthened the system by eliminating external dependencies and enabling fully offline testing. Each engine is validated independently and as part of the full pipeline, ensuring that the application delivers reliable, privacy-preserving transformations across all stages.

10. Deployment, Packaging & Future Roadmap

The *Data Anonymizer GUI* is designed as a fully offline, cross-platform desktop application. Deployment focuses on:

- simplicity
- reproducibility
- security
- portability

This post outlines the recommended deployment strategy, environment setup, packaging options, and long-term roadmap for the application. It also highlights how the transition from the original MOSTLY AI backend to the SDV-based synthetic engine simplified deployment by removing external dependencies.

10.1. Deployment Philosophy

The deployment strategy is built around three principles:

10.1.1. Offline-First Operation

All privacy transformations — cleaning, anonymization, pseudonymization, and synthetic generation — run locally.

This eliminates:

- API keys
- cloud dependencies
- network latency
- external attack surfaces

10.1.2. Reproducible Environments

The application uses a deterministic Python environment defined by:

- `requirements.txt`
- pinned versions for PySide6, pandas, SDV, and dependencies

This ensures consistent behavior across machines.

10.1.3. Minimal External Dependencies

The removal of MOSTLY AI's private SDK significantly simplified deployment.

The SDV engine is:

- pip-installable
- open-source
- compatible with Python 3.12
- fully local

This makes packaging and distribution straightforward.

10.2. Environment Setup

10.2.1. Python Version

The recommended environment is:

```
Python 3.12.x
```

10.2.2. Install Dependencies

```
pip install -r requirements.txt
```

The requirements file includes:

- PySide6
- pandas
- numpy
- scipy
- SDV
- rdt
- copulas
- matplotlib
- seaborn

10.2.3. Directory Structure

```
mostlyai_gui/  
├── core/           # backend engines  
├── gui/            # PySide6 UI components  
├── assets/         # icons, sample datasets  
├── logs/           # runtime logs  
├── output/         # cleaned/anonymized/pseudonymized/synthetic data  
├── main.py         # application entry point  
└── requirements.txt
```

This structure is intentionally simple and mirrors the MLflow Runner GUI layout.

10.3. Running the Application

From the project root:

```
python main.py
```

The GUI launches immediately and loads:

- configuration
- logging system
- all tabs
- default sample dataset (optional)

10.4. Packaging the Application

Packaging transforms the Python project into a standalone executable for distribution.

10.4.1. Recommended Tool: PyInstaller

PyInstaller is the most stable option for PySide6 applications.

10.4.2. Basic Packaging Command

```
pyinstaller --noconfirm --windowed --name "DataPrivacyWorkbench" main.spec
```

10.4.3. Spec File Considerations

The `.spec` file should include:

- PySide6 hooks
- asset folder inclusion
- output folder creation
- icon embedding

10.4.4. Output

PyInstaller produces:

```
dist/DataPrivacyWorkbench/
```

containing:

- the executable
- bundled Python runtime
- all dependencies

10.4.5. Platform Notes

- Windows: fully supported
- macOS: supported (requires signing for distribution)
- Linux: supported

Executables are **not cross-platform** — each OS requires its own build.

10.5. Security Considerations

10.5.1. Local-Only Processing

No data leaves the machine.

This is a core privacy guarantee.

10.5.2. No API Keys

The removal of MOSTLY AI eliminates:

- credential storage
- token rotation
- API misuse risks

10.5.3. Mapping Table Protection

Pseudonymization mapping tables should be:

- stored securely
- access-controlled
- optionally encrypted

10.5.4. Log Sanitization

Logs never contain:

- raw PII
- original values
- sensitive identifiers

Only transformation summaries are logged.

10.6. Performance Considerations

10.6.1. Cleaning Engine

Fast even on large datasets (100k+ rows).

10.6.2. Anonymization & Pseudonymization

Linear complexity — scales well.

10.6.3. SDV Synthetic Engine

CTGAN training is the most computationally expensive step.

Performance depends on:

- number of rows
- number of columns

- GPU availability (optional)

For typical census-style datasets (200–10,000 rows), training completes in seconds to minutes.

10.7. Extensibility & Future Roadmap

The modular architecture enables rapid expansion.

Below is the recommended roadmap.

10.7.1. Short-Term Enhancements

10.7.1.1. Differential Privacy Integration

Add noise-based privacy mechanisms:

- Laplace noise
- Gaussian noise
- DP-CTGAN

10.7.1.2. Advanced Cleaning Options

- schema validation
- duplicate detection using fuzzy matching
- outlier visualization

10.7.1.3. Enhanced Logs Tab

- color-coded log levels
- search bar
- log retention policies

10.7.2. Medium-Term Enhancements

10.7.2.1. Plugin Architecture

Allow users to add custom:

- anonymization rules
- pseudonymization strategies
- synthetic models

10.7.2.2. Multi-Dataset Workflows

Support:

- batch processing
- folder-level pipelines
- multi-table synthetic generation

10.7.2.3. Model Persistence

Allow saving/loading SDV models:

- reuse trained models
- share models internally
- version synthetic generators

10.7.3. Long-Term Enhancements

10.7.3.1. Privacy Risk Scoring

Compute:

- k-anonymity
- l-diversity
- t-closeness
- membership inference risk

10.7.3.2. Enterprise Edition

Add:

- role-based access control
- encrypted mapping table vault
- audit log export
- compliance reporting

10.7.3.3. Cloud-Optional Mode

While the application is offline-first, a future version could optionally integrate:

- remote storage
- remote model registries
- secure collaboration

This would remain opt-in to preserve privacy guarantees.

❏ 10.8. Final Remarks

Project 25 — Data Anonymizer GUI

Project 25 — the **Data Anonymizer GUI** — stands as a rare synthesis of privacy engineering, architectural clarity, and practical data-processing workflow design.

It is more than a graphical interface, more than a wrapper around pandas or SDV, and more than a convenience tool for cleaning or anonymizing datasets.

It is, in every meaningful sense, a **miniature data-privacy platform**, purpose-built for local, offline-capable, reproducible, and auditable data transformation.

The *Data Anonymizer GUI* is designed for secure, offline, and reproducible deployment.

The transition from MOSTLY AI to SDV significantly simplified packaging and eliminated external

dependencies, making the application suitable for:

- regulated industries
- air-gapped environments
- enterprise desktops
- research institutions

With a clear roadmap and modular architecture, the system is well-positioned for future enhancements in privacy engineering, synthetic data generation, and enterprise-grade governance.

What makes this project exceptional is not any single feature in isolation, but the way all components — the GUI, the cleaning engine, the anonymization and pseudonymization modules, the SDV synthetic generator, and the logging subsystem — interlock with precision.

The system is designed with a level of intentionality that is uncommon in small-scale privacy tooling: every decision, from the deterministic masking rules to the reversible pseudonymization mapping, from the SDV-based synthetic engine to the structured output directories, serves a clear purpose.

The result is a tool that is simultaneously **simple to use**, **transparent in behavior**, and **rigorous in execution**.

This extended final remarks section reflects on the project from multiple angles:

its architectural strengths, its usability, its reproducibility guarantees, its educational value, its extensibility, and its role as a bridge between data privacy practice and modern data-engineering best practices.

It also highlights the broader significance of the project in the context of contemporary data workflows, where privacy, transparency, and user-friendliness are often sacrificed in favor of automation or proprietary cloud services.

🌱 10.8.1. A Synthesis of Engineering Discipline and Practical Privacy Workflow Design

The Data Anonymizer GUI is built on a foundation of **engineering discipline**.

This discipline manifests in several ways.

10.8.1.1 Clear separation of concerns

The system is divided into five major components:

- **GUI (PySide6)** — presentation and user interaction
- **Cleaning Engine** — preprocessing and data normalization
- **Anonymization Engine** — irreversible masking
- **Pseudonymization Engine** — deterministic tokenization with mapping tables
- **Synthetic Engine (SDV CTGAN)** — generative modeling and synthetic sampling
- **Logging System** — audit trail and transparency

Each component has a single responsibility.

The GUI never manipulates data directly.

The cleaning engine never renders UI.

The pseudonymization engine never touches GUI state.

The SDV engine never interacts with file dialogs.

The logger never transforms data.

This separation ensures stability, testability, and maintainability.

10.8.1.2 Deterministic transformation logic

The anonymization and pseudonymization engines are intentionally deterministic:

- masking rules are fixed
- pseudonym tokens follow a strict format
- mapping tables ensure reversibility
- cleaning operations follow predictable heuristics

This determinism is essential for reproducibility and auditability.

10.8.1.3 Offline-first synthetic generation

The transition from MOSTLY AI's proprietary SDK to SDV's open-source CTGAN engine is a defining architectural decision.

It eliminates:

- API keys
- cloud dependencies
- network latency
- proprietary lock-in

and replaces them with:

- local training
- transparent metadata inference
- reproducible sampling
- open-source algorithms

This aligns the system with privacy-by-design principles.

10.8.1.4 Logging done right

The logging subsystem is not an afterthought — it is a core architectural pillar.

Every action is timestamped, categorized, and displayed in real time.

Users can trace:

- dataset loading
- cleaning operations
- anonymization actions
- pseudonymization mappings
- synthetic model training
- file saves

This transparency builds trust and supports compliance.



10.8.2. A Clean, Transparent, and Trustworthy Privacy Workflow

Transparency is one of the defining characteristics of the Data Anonymizer GUI.

10.8.2.1 Real-time logs

The Logs tab streams:

- cleaning summaries
- masking operations
- pseudonymization mappings
- synthetic model training progress
- file save confirmations
- warnings and errors

Nothing is hidden.

Users see exactly what the system is doing.

10.8.2.2 Explicit transformation stages

The GUI never guesses what the user wants.

Each stage is explicit:

- Clean
- Anonymize
- Pseudonymize
- Generate Synthetic Data

This eliminates ambiguity and builds confidence.

10.8.2.3 Full output visibility

Every output — cleaned, anonymized, pseudonymized, synthetic — is saved to a structured folder.

Users can inspect results at any time.

10.8.2.4 Reproducibility as a first-class citizen

The system captures:

- cleaned datasets
- anonymized datasets
- pseudonymized datasets
- mapping tables
- synthetic datasets
- logs

This ensures that every transformation is reproducible, auditable, and comparable.

In an era where privacy tooling often hides complexity behind black-box automation, the Data Anonymizer GUI makes transparency a core feature.



10.8.3. A Tool That Serves Beginners, Intermediates, and Experts

One of the most impressive aspects of the project is its ability to serve users at all skill levels.

10.8.3.1 Beginners

Beginners benefit from:

- a simple GUI
- clear transformation stages
- visual previews
- automatic logging
- safe defaults

They can perform privacy transformations without writing code.

10.8.3.2 Intermediate users

Intermediate users can:

- load their own datasets
- customize cleaning options
- inspect pseudonymization mappings
- generate synthetic datasets
- compare outputs

The GUI becomes a productivity tool.

10.8.3.3 Advanced users

Experts appreciate:

- deterministic engines
- reversible pseudonymization
- SDV CTGAN integration
- structured output directories
- audit logs

For them, the GUI is a lightweight privacy engineering platform.

This multi-level usability is extremely rare in privacy tooling.



10.8.4. A Miniature Data-Privacy Platform — Without the Complexity

The Data Anonymizer GUI is not just a GUI.

It is a **miniature data-privacy platform**, offering:

- data cleaning
- anonymization
- pseudonymization
- synthetic generation
- audit logging
- reproducibility guarantees
- structured outputs

All of this is achieved:

- without cloud services
- without proprietary SDKs
- without complex configuration
- without external dependencies

It is privacy engineering distilled to its essence.

For many teams, this tool is more than enough to establish:

- good data hygiene
- reproducible privacy workflows
- mapping table governance
- synthetic data pipelines

— without the overhead of enterprise privacy platforms.

10.8.5. Extensibility and Future-Proofing

The architecture is intentionally modular and extensible.

10.8.5.1 Extending the GUI

Future panels could include:

- differential privacy controls
- k-anonymity scoring
- dataset profiling
- multi-table synthesis
- privacy risk dashboards

The GUI is ready for growth.

10.8.5.2 Extending the engines

The engines could support:

- custom masking rules
- advanced pseudonymization strategies
- DP-CTGAN models
- constraint-based synthetic sampling

The modular design makes this straightforward.

10.8.5.3 Extending the logging system

The logger could evolve to include:

- color-coded levels
- export to JSON

- compliance-ready audit bundles

10.8.5.4 Extending the synthetic engine

SDV supports:

- GaussianCopula
- TVAE
- CopulaGAN

These can be integrated without architectural changes.

10.8.6. Educational Value and Pedagogical Strength

The Data Anonymizer GUI is an exceptional teaching tool.

10.8.6.1 Teaches privacy concepts without requiring code

Students learn:

- what anonymization is
- what pseudonymization is
- how synthetic data works
- how mapping tables function

— simply by using the GUI.

10.8.6.2 Teaches reproducibility

By saving:

- cleaned datasets
- masked datasets
- pseudonymized datasets
- synthetic datasets
- logs

students learn the importance of reproducible privacy workflows.

10.8.6.3 Teaches pipeline structure

The tab-based layout mirrors real-world privacy pipelines.

10.8.6.4 Teaches good data hygiene

The tool encourages:

- cleaning before anonymization
- deterministic transformations
- structured outputs

- audit logging

These habits are essential for professional data work.

10.8.7. A Tool That Respects the User's Environment

The Data Anonymizer GUI is designed to be:

- **local**
- **offline-capable**
- **self-contained**
- **platform-friendly**

It does not require:

- cloud accounts
- external APIs
- network access
- proprietary SDKs

This makes it ideal for:

- corporate environments
- secure research labs
- offline workshops
- educational institutions
- personal experimentation

The tool respects the user's constraints rather than imposing new ones.

10.8.8. A Transparent, Honest, and Predictable System

In a world where many privacy tools hide complexity behind automation, the Data Anonymizer GUI takes the opposite approach:

- it exposes logs
- it exposes transformations
- it exposes mapping tables
- it exposes synthetic model training
- it exposes outputs

This transparency builds trust.

Users always know:

- what is happening
- why it is happening
- where data is going
- how pseudonyms are generated
- how synthetic data is produced

There is no “magic” — only clear, predictable behavior.

10.8.9. A Foundation for Serious Privacy Work

Despite its simplicity, the Data Anonymizer GUI is not a toy.

It is a serious tool for:

- data cleaning
- anonymization
- pseudonymization
- synthetic data generation
- reproducible privacy workflows
- audit logging
- compliance support

It provides the essential infrastructure that every privacy-sensitive project needs, without the overhead of enterprise systems.

For many teams, this tool is enough to:

- standardize privacy workflows
- improve collaboration
- enforce reproducibility
- manage mapping tables
- generate safe synthetic datasets

It is a foundation that can grow with the team.

10.8.10. Final Reflection

Project 25 — the **Data Anonymizer GUI** — is a rare achievement.

It combines:

- **engineering discipline**
- **clean architecture**
- **transparent privacy workflows**
- **robust logging**
- **reproducibility**
- **user-friendly design**
- **open-source synthetic modeling**

It provides a complete, local, offline-capable privacy transformation environment that:

- guides beginners
- empowers intermediate users
- satisfies advanced users
- integrates seamlessly with SDV
- produces fully reproducible outputs
- logs everything needed for long-term governance

The project is not just a GUI — it is a **miniature data-privacy platform**, built with clarity and purpose.

It stands as a testament to what can be achieved when simplicity, transparency, and engineering rigor come together.

It is a tool that respects the user, respects the craft of data privacy, and respects the importance of reproducibility.

In a landscape crowded with overly complex privacy platforms, the Data Anonymizer GUI is a breath of fresh air:

a tool that does exactly what it promises, does it well, and does it with elegance.

11. References

1. Navoda Senavirathne / Vicenç Torra: "On the Role of Data Anonymization in Machine Learning Privacy", 2020 IEEE 19th International Conference on Trust, Security and Privacy in Computing and Communications (2020); DOI: 10.1109/TrustCom50675.2020.00093, <https://ieeexplore.ieee.org/document/9343198/authors#authors>; <https://www.datacamp.com/blog/what-is-data-anonymization>;
- Data Anonymization: <https://tryolabs.com/blog/2020/06/11/personal-data-anonymization-key-concepts--how-it-affects-machine-learning-models>; <https://mostly.ai/what-is-data-anonymization>; <https://pypi.org/project/anonym/>; <https://docs.sdv.dev/sdv>; <https://github.com/sdv-dev/sdv>; <https://pypi.org/project/sdv/1.4.0.dev1/>; <https://mostly.ai/blog/a-comparison-of-synthetic-data-vault-and-mostly-ai-part-1-single-table-scenario>; <https://medium.com/1000bytesinnovations/synthetic-data-vault-a-comprehensive-guide-62def3073844>;
- MLflow-Links: <https://mlflow.org/docs/latest/ml/>; <https://mlflow.org/docs/latest/ml/dataset/>; <https://mlflow.org/docs/latest/ml/model-registry/workflow/>;
2.   Jupyter Notebook English
3.   MLflow Runner GUI Report English
4. Tao, F., Qi, Q., Liu, A., & Kusiak, A. (2018). *Digital Twins and Cyber-Physical Systems in Manufacturing*. Engineering, 5(4);
5. A. Meister , T. Sonar: "**Numerik**", 1st Ed. Springer-Spektrum (2019); S. Chapra, R. Canale: "**Numerical Methods for Engineers**", Mcgraw-Hill, 6th Edition (2010).
6. J. Kilty, A. M. McAllister: "**Mathematical Modeling and Applied Calculus**", 1st Ed. Oxford University Press (2018).
7. U. Kockelkorn: "**Statistik für Anwender**", 1st Ed. Springer (2012), s. chapters 7 - 8.
8. Robert H. Shumway, David S. Stoffer: "**Time Series Analysis and Its Applications with R Examples**", Springer (2011).
9. Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, Jonathan Taylor: "**An Introduction to Statistical Learning with Applications in Python**", Springer (2023).
10. Cornelis W. Oosterlee, Lech A. Grzelak: "**Mathematical Modeling and Computation in Finance with Exercises and Python and MATLAB Computer Codes**", World Scientific (2020).

11. Lee, J., Bagheri, B., & Kao, H. (2015). *A Cyber-Physical Systems architecture for Industry 4.0-based manufacturing systems*. Manufacturing Letters;
12. Richard Szeliski: "**Computer Vision - Algorithms and Applications**", Springer (2022).
13. Anthony Scopatz, Kathryn D. Huff: "**Effective Computation in Physics - Field Guide to Research with Python**", O'Reilly Media (2015).
14. Alex Gezerlis: "**Numerical Methods in Physics with Python**", Cambridge University Press (2020).
15. Gary Hutson, Matt Jackson: "**Graph Data Modeling in Python. A practical guide**", Packt-Publishing (2023).
16. Hagen Kleinert: "**Path Integrals in Quantum Mechanics, Statistics, Polymer Physics, and Financial Markets**", 5th Edition, World Scientific Publishing Company (2009).
17. Peter Richmond, Jurgen Mimkes, Stefan Hutzler: "**Econophysics and Physical Economics**", Oxford University Press (2013).
18. A. Coryn , L. Bailer Jones: "**Practical Bayesian Inference A Primer for Physical Scientists**", Cambridge University Press (2017).
19. Avram Sidi: "**Practical Extrapolation Methods - Theory and Applications**", Cambridge university Press (2003).
20. Volker Ziemann: "**Physics and Finance**", Springer (2021).
21. Zhi-Hua Zhou: "**Ensemble methods, foundations and algorithms**", CRC Press (2012).
22. B. S. Everitt, et al.: "**Cluster analysis**", Wiley (2011).
23. Lior Rokach, Oded Maimon: "**Data Mining With Decision Trees - Theory and Applications**", World Scientific (2015).
24. Bernhard Schölkopf, Alexander J. Smola: "**Learning with kernels - support vector machines, regularization, optimization and beyond**", MIT Press (2009).
25. Johan A. K. Suykens: "**Regularization, Optimization, Kernels, and Support Vector Machines**", CRC Press (2014).
26. Sarah Depaoli: "**Bayesian Structural Equation Modeling**", Guilford Press (2021).
27. Rex B. Kline: "**Principles and Practice of Structural Equation Modeling**", Guilford Press (2023).
28. Ekaterina Kochmar: "**Getting Started with Natural Language Processing**", Manning (2022).
29. Jakub Langr, Vladimir Bok: "**GANs in Action**", Computer Vision Lead at Founders Factory (2019).
30. David Foster: "**Generative Deep Learning**", O'Reilly(2023).
31. Rowel Atienza: "**Advanced Deep Learning with Keras: Applying GANs and other new deep learning algorithms to the real world**", Packt Publishing (2018).
32. Josh Kalin: "**Generative Adversarial Networks Cookbook**", Packt Publishing (2018).
33. Thomas Haslwanter: "**Hands-on Signal Analysis with Python: An Introduction**", Springer (2021).
34. Jose Unpingco: "**Python for Signal Processing**", Springer (2023).
35. R. K. Burdick, C. M. Borror, D. C. Montgomery: "**Design and Analysis of Gauge R&R Studies**", 1st Ed. SIAM (2005); S. H. Derakhshan , C. V. Deutsch: "**Numerical Integration of Bivariate Gaussian Distribution**", Paper 405, CCG Anual Report 13 (2011).
36. C. Paar, J. Pelzl: "**Understanding Cryptography**", Springer (2010); H. Delfs, H. Knebl: "**Introduction to Cryptography**", 3rd Ed. Springer (2015); J. Katz, Y. lindell: "**Introduction to Modern Cryptography**", 2nd Ed, CRC Press (2015); O. Goldreich: "**Foundations of Cryptography**", Cambridge University Press (2008); J. P. Aumasson: "**Serious Cryptography**", no starch press (2018).
37. J. Berk, P. DeMarzo: „**Corporate Finance**“, 6th Ed., Pearson (2023); R. W. Melicher, E. A. Norton: "**Introduction to Finance**", 16th Ed. WILEY (2017); Anatoly B. Schmidt: "**Quantitative Finance for Physicists: An Introduction**", 1st Ed. Academic Press (2005); Alex Backwell: "**An Intuitive Introduction to Finance and Derivatives: Concepts, Terminology and Models**", 1st Ed, Springer (2023); Michael

- Isichenko: **"Quantitative Portfolio Management: The Art and Science of Statistical Arbitrage"**, 1st Ed., Springer (2021); John H. Cochrane: **"Asset Pricing"**, Revised Ed., Princeton University Press (2005); Antti Ilmanen: **"Expected Returns: An Investor's Guide to Harvesting Market Rewards"**, 1st Ed., WILEY (2011); Steven E. Shreve: **"Stochastic Calculus for Finance I & II"**, 1st Ed., Springer (2004); Andrew Pole: **"Statistical Arbitrage: Algorithmic Trading Insights and Techniques"**, 1st Ed., WILEY (2007); Mark S. Joshi: **"The Concepts and Practice of Mathematical Finance"**, 2nd Ed., Cambridge University Press (2008); Kaggle-link: competition-documentation: <https://www.kaggle.com/competitions/drw-crypto-market-prediction>.
38. R. Nystrom: **"Game Programming Patterns"**, 1st Ed. genever benning (2014); A. A. Stepanov, D. E. Rose: **"From Mathematics to Generic Programming"**, 1st Ed. Addison-Wesley (2015);
39. E. Parzen: **"Stochastic Processes"**, 3rd Ed. Dover Publications (2015); S. Aloorravi: **"Metaprogramming with Python"**, 1st Ed. Packt (2022); B. Klein, P. Klein: **"Funktionale Programmierung mit Python"**, Hanser (2025); K. Webel, D. Wied: **"Stochastische Prozesse"**, 2. Auflage Springer (2016); L. Held: **"Methoden der statistischen Inferenz"**, 1. Auflage Spektrum (2008); E. Cinlar: **"Stochastic Processes"**, Dover (2013); N. Bäuerle, U. Rieder: **"Finanzmathematik in diskreter Zeit"**, Springer-Spektrum (2017); M. Albrecht, R. Maurer: **"Investment- und Risikomanagement"**, 3. Auflage, Schäffer Poeschel (2008); N. H. Bingham, R. Kiesel: **"Risk Neutral Valuation: Pricing and Hedging of Financial Derivatives"**, 2. Auflage Springer (2004); T. Björk: **"Arbitrage Theory in Continuous Time"**, 3rd Ed. Oxford University Press (2009); N. J. Cutland, A. Roux: **"Derivative Pricing in Discrete Time"**, Springer (2013); F. Delbaen, W. Schachermayer: **"The Mathematics of Arbitrage"**, Springer (2006); R. J. Elliott, P. E. Kopp: **"Mathematics of Financial Markets"**, 2nd Ed. Springer (2005); H. Föllmer, A. Scheid: **"A Stochastic Finance: An Introduction in Discrete Time"**, 3rd Ed. de Gruyter (2011); J. C. Hull: **"Options, Futures and Other Derivatives"**, 8th Ed. Pearson (2011); J. Kremer: **"Einführung in die diskrete Finanzmathematik"**, Springer (2005); D. Lamberton, B. Lapeyre: **"Introduction to Stochastic Calculus Applied to Finance"**, Chapman & Hall (2007); D. G. Luenberger: **"Investment Science"**, Oxford University Press (1998); S. R. Pliska: **"Introduction to Mathematical Finance: Discrete Time Models"**, Blackwell (2000); A. N. Shiryaev: **"Essentials of Stochastic Finance"**, World Scientific (2001); S. E. Shreve: **"Stochastic Calculus for Finance I: The Binomial Asset Pricing Model"**, Springer (2004); J. Kremer: **"Portfoliotheorie, Risikomanagement und die Bewertung von Derivaten"**, Springer (2011); L. Rüschendorf: **"Mathematical Risk Analysis"**, Springer (2013).
40. A. Becker: **"Kalman Filter - From the Ground Up"**, 1st Ed. private publication (2023); K. Triantafyllopoulos: **"Bayesian Inference of State Space Models"**, 1st Ed. Springer (2021); P. Zarchan, H. Musoff: **"Fundamentals of Kalman Filtering: A Practical Approach"**, 3rd Ed. AIAA (2009); A. Sidi: **"Vector Extrapolation Methods with Applications"**, 1st Ed. SIAM (2019); C. Brezinski, M. R. Zaglia: **"Extrapolation Methods - Theory and Practice"**, 2nd Ed. North-Holland (2002); C. Gardiner, P. Zoller: **"Quantum Noise: A Handbook of Markovian and Non-Markovian Quantum Stochastic Methods with Applications to Quantum Optics"**, 3rd Ed. Springer (2004); K. Kendre: **"Machine Learning for Quantum Noise Reduction"**, <https://arxiv.org/abs/2509.16242> (2025); D. C. Marinescu, G. M. Marinescu: **"Classical and Quantum Information"**, 1st Ed. Academic Press (2012); Liao, H et al.: **"Machine Learning for Practical Quantum Error Mitigation"**, arXiv:2309.17368v2 (2024), <https://arxiv.org/pdf/2309.17368>; Streamlit: <https://streamlit.io/>; Mitiq-package: <https://quantum-journal.org/papers/q-2022-08-11-774/>, <https://arxiv.org/abs/2009.04417>; Extrapolation packages: <https://pypi.org/project/extrapolation/>
41. A. Koop, H. Moock: **"Lineare Optimierung - Eine anwendungsorientierte Einführung in Operations Research"**, 1st Ed. Spektrum (2008); G. B. Dantzig, M. N. Thalpa: **"Linear Programming 1: Introduction"**, 1st Ed. Springer (1997) & **"Linear Programming 2: Theory and Extensions"**, 1st Ed.

- Springer (2003); H. S. Kasana, K. D. Kumar: "**Introductory Operations Research, Theory and Applications**", 1st Ed. Springer (2004); D. G. Luenberger: "**Linear and Nonlinear Programming**", 2nd Ed. Kluwer (2004); R. J. Boucherie, A. Braaksma, H. Tijms: "**Operations Research - Introduction to Models and Methods**", 1st Ed. World Scientific (2022); A. J. King, S. W. Wallace: "**Modeling with Stochastic Programming**", 2nd Ed. Springer (2024); J. O. Royset, R. J.-B. Wets: "**An Optimization Primer**", 1st Ed. Springer (2021); cvxpy package: <https://www.cvxpy.org/>, <https://pypi.org/project/cvxpy/>; py-packages for operations research: <https://wiki.python.org/moin/PythonForOperationsResearch>
42. (Py-)tesseract package: <https://github.com/tesseract-ocr/tesseract>, <https://pypi.org/project/pytesseract/>, <https://builtin.com/data-science/python-ocr>, <https://www.analyticsvidhya.com/blog/2024/04/ocr-libraries-in-python/> and [UB Mannheim builds](#).
43. **Chip Huyen**, *AI Engineering: Building Applications with Foundation Models*, 1st Edition, O'Reilly Media, 2025; **Michael Lanham**, *AI Agents in Action*, 1st Edition, Manning Publications, 2025; **Melanie Mitchell**, *Artificial Intelligence: A Guide for Thinking Humans*, 1st Edition, Pelican Books, 2019; **Brian Christian & Tom Griffiths**, *Algorithms to Live By: The Computer Science of Human Decisions*, 1st Edition, Henry Holt and Company, 2016; **Ray Kurzweil**, *The Singularity Is Nearer: When We Merge with AI*, 1st Edition, Viking, 2024; OpenWeatherMap: <https://openweathermap.org/>, HuggingFace: <https://huggingface.co/>,
44. J. Frochte: "Finite-Elemente-Methode", Hanser 1st Ed.(2016); D. Gross, W. Hauger, J. Schröder: "Technische Mechanik 1-3", 15th Ed. Springer (2024); FEM-packages (Python): <https://pypi.org/project/scikit-fem/>, <https://sfepy.org/doc-devel/index.html>, https://getfem-examples.readthedocs.io/en/latest/demo_unit_disk.html, <https://github.com/mlp6/fem>. LLM vs LRM: <https://www.aryaxai.com/article/llm-vs-lrm-vs-lam-understanding-the-future-of-language-based-ai-systems>, <https://magazine.sebastianraschka.com/p/understanding-reasoning-llms>
45. Grieves, M. (2015). *Digital Twin: Manufacturing Excellence through Virtual Factory Replication.*; Rasheed, A., San, O., & Kvamsdal, T. (2020). *Digital Twin: Values, Challenges and Enablers*. IEEE Access.; Jones, D., Snider, C., Nassehi, A., Yon, J., & Hicks, B. (2020). *Characterising the Digital Twin: A systematic literature review*. CIRP Journal of Manufacturing Science and Technology; Tao, F., & Zhang, M. (2017). *Digital Twin Shop-Floor: A new shop-floor paradigm towards smart manufacturing*. IEEE Access; Glaessgen, E., & Stargel, D. (2012). *The Digital Twin Paradigm for Future NASA and U.S. Air Force Vehicles.*; Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press; Molnar, C. (2020). *Interpretable Machine Learning.*; Microsoft. *PySide6 Documentation.*: <https://pypi.org/project/PySide6/>; Apache Arrow. *Parquet File Format Specification.*: <https://arrow.apache.org/docs/python/parquet.html>; NumPy Developers. *NumPy Reference Guide.*: <https://numpy.org/doc/stable/reference/>; Matplotlib Developers. *Matplotlib Plotting Library.*: <https://matplotlib.org/>;
46. Navoda Senavirathne / Vicenç Torra: "On the Role of Data Anonymization in Machine Learning Privacy", 2020 IEEE 19th International Conference on Trust, Security and Privacy in Computing and Communications (2020); DOI: 10.1109/TrustCom50675.2020.00093, <https://ieeexplore.ieee.org/document/9343198/authors#authors>; <https://www.datacamp.com/blog/what-is-data-anonymization>; <https://tryolabs.com/blog/2020/06/11/personal-data-anonymization-key-concepts--how-it-affects-machine-learning-models>; <https://mostly.ai/what-is-data-anonymization>; <https://pypi.org/project/anonym/>.